

Human Brain-Wide Activation of Sleep Rhythms

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The study reports **valuable** findings from a very rich EEG-fMRI dataset, including 107 participants, which was collected during nocturnal naps. Using overall **solid** methods, the authors link activity in memory-related brain regions (e.g., hippocampus, thalamus, and medial prefrontal cortex), and their functional connectivity to the occurrence of canonical sleep rhythms (spindles and slow oscillations) in non-rapid eye movement sleep. This work will be of broad interest to sleep and memory researchers and beyond.

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Abstract

During sleep, our brain undergoes highly synchronized activity, orchestrated by distinct neural rhythms. Little is known about the associated brain activation during these sleep rhythms, and even less about their functional implications. In this study, we investigated the brain-wide activation underlying human sleep rhythms by employing simultaneous electroencephalography (EEG) and functional magnetic resonance imaging (fMRI) in 107 participants during nocturnal nap (first half of the night). We identified robust coupling between slow oscillations (SOs) and fast spindles during deep non-rapid eye movement (NREM) sleep (N2/3 stages), with spindle peaks consistently occurring just before the SO UP-state. This SO-spindle coupling was linked to elevated activation in both the thalamus and hippocampus, alongside increased functional connectivity from the hippocampus to the thalamus and from the thalamus to the medial prefrontal cortex (mPFC). An open-ended cognitive state decoding analysis suggested that these activations may relate to episodic memory processes, yet were distinct from task-related networks. Together, these findings highlight the thalamus as a key coordinator of hippocampal-cortical communication during sleep and provide new insights into the mechanisms by which synchronized sleep rhythms may support memory consolidation.

Introduction

During sleep, the human brain engages in highly synchronized activity driven by distinct neural rhythms. In non-rapid eye movement (NREM) sleep, the emergence and coordination of slow oscillations (SOs) and sleep spindles are key features (Hahn et al., 2020; Helfrich et al., 2018; Ngo, Fell, & Staresina, 2020; Schreiner et al., 2022; Staresina et al., 2015; Staresina et al., 2023). However, the brain activation patterns associated with these rhythms, and especially their coordination, remain largely unexplored, and their functional implications are even less understood.

SOs (~1 Hz) are characterized by fluctuations in neuronal membrane potential between periods of silence ('hyperpolarization', or 'DOWN' state) and excitation ('depolarization', or 'UP' state) (Amzica & Steriade, 2002; Steriade, McCormick, & Sejnowski, 1993). These oscillations originate in neocortical regions, particularly the medial prefrontal cortex (mPFC), during NREM sleep (Dang-Vu et al., 2008; Massimini et al., 2004; Nir et al., 2011). The depolarization phase of SOs is believed to trigger sleep spindles in the thalamus, producing 11–16 Hz oscillations (Fernandez & Lüthi, 2020; Mak-McCully et al., 2017). Thalamic spindles tend to align with the excitable UP-states of SOs and can synchronize with hippocampal ripples (Buzsáki, 2015; Helfrich et al., 2019; Joo & Frank, 2018; Ngo, Fell, & Staresina, 2020). Rodent optogenetics studies have shown that only thalamic spindles phase-locked to the UP-state of cortical SOs enhance memory consolidation, while out-of-phase spindles do not (Latchoumane et al., 2017). This finding is consistent with intracranial evidence from human epilepsy patients, suggesting the thalamus acts as a key relay between the hippocampus and cortex, coordinating their interaction (Coulon, Budde, & Pape, 2012; Ferraris et al., 2021; Schreiner et al., 2022; Staresina et al., 2015).

The triple coupling of SOs, spindles, and ripples is believed to facilitate memory consolidation by synchronizing neuronal activity across brain regions and replaying memory traces from the hippocampus to the neocortex (Diekelmann & Born, 2010; Latchoumane et al., 2017; Liu et al., 2022; Singh, Norman, & Schapiro, 2022; Staresina et al., 2015; Staresina et al., 2023), possibly through relay from thalamic spindles. However, the exact mechanisms of this inter-regional communication during coordinated sleep rhythms remain unclear. Understanding the brain-wide activation associated with these rhythms is crucial, as it reveals how information flows among brain regions, especially during the critical time windows when these rhythms align.

Investigating these sleep-related neural processes in humans is challenging because it requires tracking transient sleep rhythms while simultaneously assessing their widespread brain activation. Recent advances in simultaneous EEG-fMRI techniques provide a unique opportunity to explore these processes. EEG allows for precise event-based detection of neural signal, while fMRI provides insight into the broader spatial patterns of brain activation and functional connectivity (Horovitz et al., 2008; Huang et al., 2024; Laufs, 2008; Laufs, Walker, & Lund, 2007; Schabus et al., 2007; Spoormaker et al., 2010). Previous EEG-fMRI studies on sleep have focused on classifying sleep stages or examining the neural correlates of specific waves (Bergmann et al., 2012; Caporro et al., 2012; Czisch et al., 2009; Fogel et al., 2017; Hale et al., 2016; Ilhan-Bayrakci et al., 2022; Moehlmán et al., 2019; Picchioni et al., 2011). These studies have generally reported that slow oscillations are associated with widespread cortical and subcortical BOLD changes, whereas spindles elicit activation in the thalamus, as well as in several cortical and paralimbic regions. Although these findings provide valuable insights into the BOLD correlates of sleep rhythms, they often do not employ sophisticated temporal modeling (Huang et al., 2024), to capture the dynamic interactions between different oscillatory events, e.g., the coupling between SOs and spindles.

Understanding sleep rhythm-related brain activation is also essential for uncovering its potential cognitive functions. Since performing cognitive tasks during sleep is limited, most neuroimaging studies on human sleep rely on the targeted memory reactivation (TMR) paradigm, where participants are re-exposed to sensory cues during sleep to trigger associated memory reactivation (Cousins et al., 2016 [↗](#); Hu et al., 2020 [↗](#); Oudiette & Paller, 2013 [↗](#); Siefert et al., 2024 [↗](#)). For instance, fMRI studies have shown significant hippocampal activation in response to odour re-exposure during sleep (Rasch et al., 2007 [↗](#)), while EEG studies suggest that this reactivation is more likely during the UP-state of SOs (Cousins et al., 2016 [↗](#); Göldi et al., 2019 [↗](#); Xia et al., 2023 [↗](#)). However, while TMR provides insight into externally induced memory processing during sleep, it offers limited understanding of the endogenous neural mechanisms underlying natural sleep rhythms.

To address this gap, our study investigates brain-wide activation and functional connectivity patterns associated with SO-spindle coupling, and employs a cognitive state decoding approach (Margulies et al., 2016 [↗](#); Yarkoni et al., 2011 [↗](#))—albeit indirectly—to infer potential cognitive functions. In the current study, we used simultaneous EEG-fMRI recordings during nocturnal naps (detailed sleep staging results are provided in the Methods and **Table S1** [↗](#)) in 107 participants. Although directly detecting hippocampal ripples using scalp EEG or fMRI is challenging, we expected that hippocampal activation in fMRI would coincide with SO-spindle coupling detected by EEG, given that SOs, spindles, and ripples frequently co-occur during NREM sleep. We also anticipated a critical role of the thalamus, particularly thalamic spindles, in coordinating hippocampal-cortical communication.

We found significant coupling between SOs and spindles during NREM sleep (N2/3), with spindle peaks occurring slightly before the SO peak. This coupling was associated with increased activation in both the thalamus and hippocampus, with functional connectivity patterns suggesting thalamic coordination of hippocampal-cortical communication. These findings highlight the key role of the thalamus in coordinating hippocampal-cortical interactions during human sleep and provide new insights into the neural mechanisms underlying sleep-dependent brain communication. A deeper understanding of these mechanisms may contribute to future neuromodulation approaches aimed at enhancing sleep-dependent cognitive function and treating sleep-related disorders.

Results

Simultaneous EEG-fMRI data during nocturnal sleep were collected from 138 healthy subjects (age, 22.56 ± 0.30 ; 81 females). After excluding subjects due to excessive head motion (mean framewise displacement > 0.5) or insufficient scanning sleep duration (< 10 min), 107 participants (age, 22.40 ± 0.34 ; 63 females) were retained for subsequent analysis, with an average scanning time of sleep of 3.01 ± 0.14 hours (see Methods and **Table S1** [↗](#)). Since NREM sleep predominantly occurs in early sleep (Carskadon & Dement, 2005 [↗](#)), we conducted our scans during the first half of the night to ensure sufficient coverage of NREM sleep, this is also the period when sleep rhythms are most consistently observed and studied in relation to memory consolidation (Maingret et al., 2016 [↗](#); Molle et al., 2011 [↗](#); Schreiner et al., 2021 [↗](#); Staresina et al., 2015 [↗](#); Staresina et al., 2023 [↗](#)).

Sleep stages and sleep rhythms

The primary objective of the EEG analysis was to examine the intrinsic characteristics of sleep rhythms and to identify the timing of these SO and spindle events for subsequent fMRI analyses.

We first removed MRI gradient noise from the EEG signal (detailed in Methods, see **Fig. S1** [↗](#)) and then applied an automated sleep staging algorithm (Vallat & Walker, 2021 [↗](#)). The staging results were manually reviewed by two sleep experts to ensure the accurate classification of each sleep

stages (**Fig. 1b**). Upon validation, we confirmed that N2/3 stages predominated in the dataset, with a significantly higher proportion of N2/3 sleep ($75.77\% \pm 1.50\%$) compared to N1 ($14.57\% \pm 1.06\%$, $t_{(106)} = 25.04$, $p < 1e-4$, paired-samples t test) and REM sleep ($9.66\% \pm 0.90\%$, $t_{(106)} = 29.60$, $p < 1e-4$, **Fig. 1d**, **Table S1**).

Each sleep stage is characterized by distinct spectral properties and rhythmic waveforms, serving as physiological markers (**Fig. 1c**). As a sanity check, we compared SO and spindle amplitudes and densities across stages. Both were significantly higher during N2/3 sleep (SO: $25.59 \pm 1.49 \mu\text{V}$, spindle: $7.39 \pm 0.27 \mu\text{V}$) compared to N1 (SO: $18.87 \pm 1.08 \mu\text{V}$, spindle: $6.72 \pm 0.25 \mu\text{V}$) and REM (SO: $16.67 \pm 1.14 \mu\text{V}$, spindle: $6.23 \pm 0.23 \mu\text{V}$, all $p < 1e-4$, **Fig. 1e**, **Fig. S2**). This aligns with the established understanding that N2/3 stages feature more pronounced slow-wave activity and robust spindles (Gaillard & Blois, 1981). Furthermore, we found that the amplitude of the SO DOWN-state ($13.62 \pm 0.67 \mu\text{V}$) was significantly higher than the UP-state ($12.98 \pm 0.66 \mu\text{V}$, $t_{(106)} = 4.82$, $p < 1e-4$), consistent with previous work (Cash et al., 2009; Dang-Vu et al., 2008). Based on this, we used the DOWN-state to lock the timing of SOs for subsequent EEG-informed fMRI analysis. Descriptive information on sleep rhythms during these stages for all 107 subjects can be found in **Table S2-S4**.

SO-spindle coupling

SO-spindle coupling is considered important for sleep-dependent memory consolidation. In current study, we found that this coupling primarily occurs during N2/3 sleep stages (2.46 ± 0.06 times/min), significantly more than in N1 (2.23 ± 0.09 times/min, $t_{(106)} = 2.81$, $p = 0.0059$) and REM stages (2.32 ± 0.09 times/min, $t_{(106)} = 1.99$, $p = 0.0492$, **Fig. 2b**, **Table S2-S4**), consistent with previous findings (Ngo et al., 2013; Staresina et al., 2015). After extracting all EEG epochs where SO-spindle coupling occurred, we analysed their spectral and phase characteristics. The spindles were most likely to occur slightly before the UP state peak of SOs (**Fig. 2a, e**), aligning with results from both animal studies (Maingret et al., 2016) and human research (Staresina et al., 2015). In our data, this pattern was consistent across subjects (**Fig. 2d**, Rayleigh test: $z = 9.51$, $p < 1e-4$), with the peak of the spindle aligned at an SO phase of $-41.61 \pm 0.86^\circ$ (the SO UP state peak is 0°).

Interestingly, spindle events coupled with SOs had higher amplitudes ($7.69 \pm 0.26 \mu\text{V}$) compared to those not coupled ($7.33 \pm 0.25 \mu\text{V}$, $t_{(106)} = 4.65$, $p < 1e-4$, **Fig. 2c**, left). Similarly, SO events coupled with spindles exhibited higher amplitudes ($26.28 \pm 1.34 \mu\text{V}$) than those not coupled ($24.56 \pm 1.33 \mu\text{V}$, $t_{(106)} = 6.86$, $p < 1e-4$, **Fig. 2c**, right). These findings suggest coordinated neural processing during SO-spindle coupling. Next, we explore the brain-wide activation associated with these sleep rhythms.

Sleep rhythms associated brain activation

To investigate brain-wide activity during sleep rhythms, we modeled the fMRI blood oxygen level-dependent (BOLD) signal based on EEG-identified timing of SOs and spindle events during N2/3 sleep stages. The design matrix included the main effects of SOs, spindles, and their coupling (**Fig. 3a**, detailed in Methods). This method is validated in our previous work to effectively capture replay-aligned brain activation during rest (Huang et al., 2024). We found that SOs were associated with positive activation in the thalamus (ROI analysis, $t_{(106)} = 2.38$, $p = 0.0096$, one-sample t test) and significant deactivation in the neocortex (**Fig. 3b**, see also **Fig. S5**), particularly in the default mode network (DMN), which includes the mPFC and posterior cingulate cortex (PCC), consistent with previous research (Dang-Vu et al., 2008; Picchioni et al., 2011; Tagliazucchi et al., 2013). This deactivation of the neocortex, particularly the DMN, at the trough of SOs is intriguing, given that the DMN is thought to encode the internal model of the world or cognitive map (Baldassano, Hasson, & Norman, 2018; Constantinescu, O'Reilly, & Behrens, 2016; Park, Miller, & Boorman, 2021).

Fig. 1

Sleep stages and sleep rhythms in 107 subjects.

a, Sleep rhythms and task schematic. Subjects slept first half of the night with simultaneous EEG-fMRI recordings. Since detecting hippocampal ripples directly from scalp EEG is challenging, our focus was on capturing SOs, spindles, and their couplings. Regions of interest (ROIs) are colour-coded: green for the thalamus (spindle), purple for the mPFC (SOs), and orange for the hippocampus (ripples). **b**, Sleep staging and EEG spectrogram. N2/3 sleep stages (red line) were initially identified using an offline automatic sleep staging algorithm (Vallat & Walker, 2021) and then manually validated. **(c)** Schematic of EEG data across different sleep stages, using preprocessed data from the C3 electrode. **(d)** Proportion of each sleep stage in the dataset. **(e)** Amplitudes (μV) of SOs (left) and spindles (right) across sleep stages. The SO amplitudes used here are from 0.16-1.25 Hz band EEG data, and the spindle amplitudes used here are from 12-16 Hz band EEG data. Each dot represents an individual participant. Error bars indicate SEM. *** $p < 0.001$.

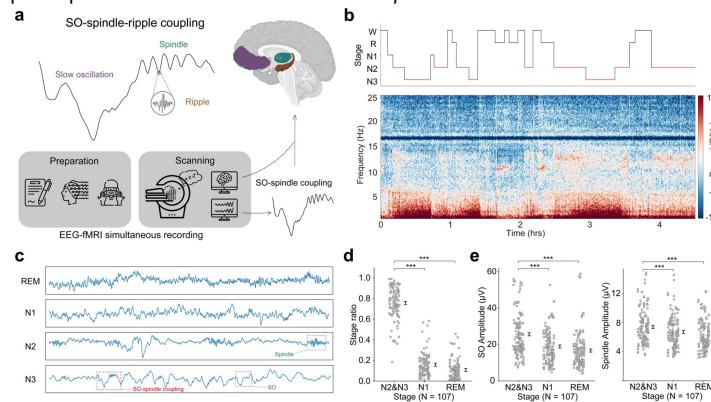
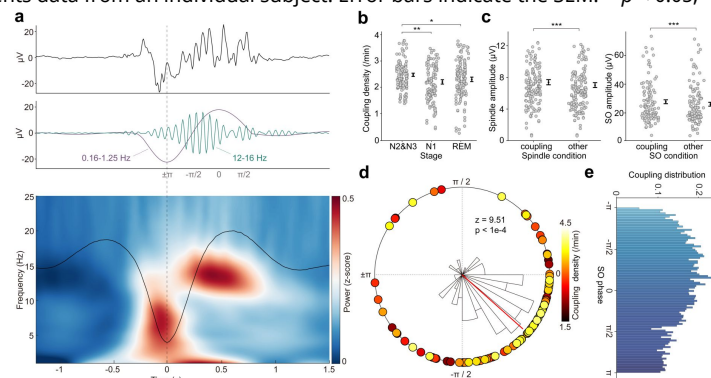


Fig. 2

Sleep rhythms and SO-spindle coupling.

a, The SO-spindle coupling in the temporal frequency domain. The upper two rows illustrate the spindle (12-16 Hz) phase-locked in the transition to UP-state of SO (0.16-1.25 Hz). The bottom row shows the averaged temporal frequency pattern across all instances of SO-spindle coupling and over all subjects. **b**, SO-spindle coupling density across different sleep stages. **c**, Differences between coupled and uncoupled sleep rhythms. The left panel shows the difference in amplitude between spindles coupled with SOs (Coupling) and spindles not coupled with SOs (Other). The right panel displays the difference in amplitude between SOs coupled with spindles (Coupling) and SOs not coupled with spindles (Other). **d**, Phase modulation of SO-spindle coupling. Spindle peaks cluster slightly before the UP-state peak of SO (i.e., 0°), where $-\pi/2$ reflects the transition from DOWN to UP-state. The histogram represents the distribution of coupling directions across all subjects, with the red line showing the mean. Coupling phases for each subject are plotted on the circle, with coupling strength color-coded. **e**, Distribution of spindle peaks on the SO phase during all SO-spindle coupling events across participants. The distribution is represented by a probability density function, and the density is evaluated at 100 equally spaced points covering the data range. Each dot represents data from an individual subject. Error bars indicate the SEM. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.



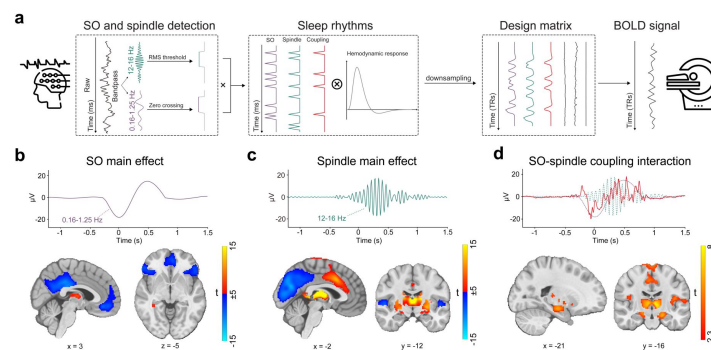


Fig. 3

Brain-wide activation associated with sleep rhythms.

a, Simultaneous EEG-fMRI analysis framework for detecting brain-wide activation during sleep rhythms. Detected SOs, spindles, and their coupling were convolved with the hemodynamic response function (HRF) and downsampled to match fMRI temporal resolution. These events formed the design matrix for the general linear model (GLM) analysis of fMRI activity during sleep, linking the EEG-derived timing of sleep rhythms to the corresponding brain responses in fMRI. **b**, Brain-wide activation associated with SOs. The upper row illustrates SOs, and the lower row shows the fMRI activation pattern during SO events, whole-brain family-wise error (FWE) corrected at the cluster level ($p < 0.05$) with a cluster-forming voxel threshold of $p < 0.001$. **c**, Brain-wide activation associated with spindles. Same as panel b, but for spindle events. **d**, Brain-wide activation associated with SO-spindle coupling (compared to non-coupling events).

Spindles were correlated with positive activation in the thalamus (ROI analysis, $t_{(106)} = 15.39$, $p < 1e-4$), the anterior cingulate cortex (ACC), and the putamen, alongside deactivation in the DMN (**Fig. 3c**). Notably, SO-spindle coupling was linked to significant activation in both the thalamus (ROI analysis, $t_{(106)} = 3.38$, $p = 0.0005$) and the hippocampus (ROI analysis, $t_{(106)} = 2.50$, $p = 0.0070$, **Fig. 3d**). However, no decrease in DMN activity was found during SO-spindle coupling, and DMN activity during SO was significantly lower than during coupling (ROI analysis, $t_{(106)} = -4.17$, $p < 1e-4$). For more detailed activation patterns, see **Table S5-S7**. We also varied the threshold used to detect SO events to assess its effect on hippocampal activation during SO-spindle coupling and observed that hippocampal activation remained significant when the percentile thresholds for SO detection ranged between 71% and 80% (see **Fig. S6**).

To understand the functional implications of these sleep rhythm-related brain activation, we performed open-ended cognitive state decoding using the NeuroSynth database (Yarkoni et al., 2011), examining associations between topic terms and regions of interest derived from the activation patterns of the sleep rhythms. Topic terms were ranked by their z-scores across the rhythms to assess overall functional relevance (see Methods for details). For positive activation patterns (**Fig. S7a**), SOs were associated with declarative ($z = 6.31$, $p < 1e-4$) and episodic memory ($z = 5.08$, $p < 1e-4$), while spindles aligned most with working memory ($z = 4.08$, $p < 1e-4$). SO-spindle coupling showed the strongest similarity to episodic ($z = 3.66$, $p = 0.0003$) and declarative memory ($z = 2.21$, $p = 0.0271$). For negative (deactivation) patterns (**Fig. S7b**), SOs were linked to declarative memory ($z = 7.61$, $p < 1e-4$), while spindles resembled task-related activations ($z = 7.05$, $p < 1e-4$). SO-spindle coupling had the highest similarity to task-related ($z = 9.40$, $p < 1e-4$) and working memory ($z = 9.15$, $p < 1e-4$). These findings suggest that internal memory-related regions are more likely to reactivate during SO-spindle coupling, whereas task-related activations, like working memory, are less prominent.

Functional connectivity during SO-spindle coupling

Previous studies have indicated that the hippocampal-cortical dialogue during sleep memory consolidation relies on the thalamus (Coulon, Budde, & Pape, 2012; Ferraris et al., 2021; Latchoumane et al., 2017). To explore this in the human brain, we examined how functional connectivity (FC) changes when spindles couple (versus not) with SOs during sleep, using psychophysiological interaction (PPI) analysis (Friston et al., 1997). Building on the previous GLM (**Fig. 3a**), this model incorporated two additional regressors: the BOLD signal of the ROI seed and its interaction with SO-spindle events (**Fig. 4a**, detailed in Methods).

We first examined whole-brain FC (physiological term) in three ROIs: the hippocampus (bilateral, anatomically defined), thalamus (bilateral, anatomically defined), and mPFC (bilateral, functionally defined). The hippocampus-based FC was mainly connected with the DMN, particularly the mPFC (**Fig. 4b**). The thalamus-based FC was primarily linked to the basal ganglia and ACC (**Fig. 4c**), and the mPFC-based FC showed strong connections with the hippocampus and PCC (**Fig. 4d**).

Crucially, the interaction between FC and SO-spindle coupling revealed that only the functional connectivity of hippocampus $\rightarrow$ thalamus (ROI analysis, $t_{(106)} = 1.86$, $p = 0.0328$) and thalamus $\rightarrow$ mPFC (ROI analysis, $t_{(106)} = 1.98$, $p = 0.0251$) significantly increased during SO-spindle coupling, with no significant changes in all other pathways (**Fig. 4e**). We also conducted PPI analyses for the other two events (SOs and spindles), and neither yielded significant connectivity changes in the three ROIs, as all failed to survive whole-brain FWE correction at the cluster level ($p < 0.05$). Together, these findings suggest that the thalamus, likely via spindles, coordinates hippocampal-cortical communication selectively during SO-spindle coupling, but not isolated SOs or spindle events alone.

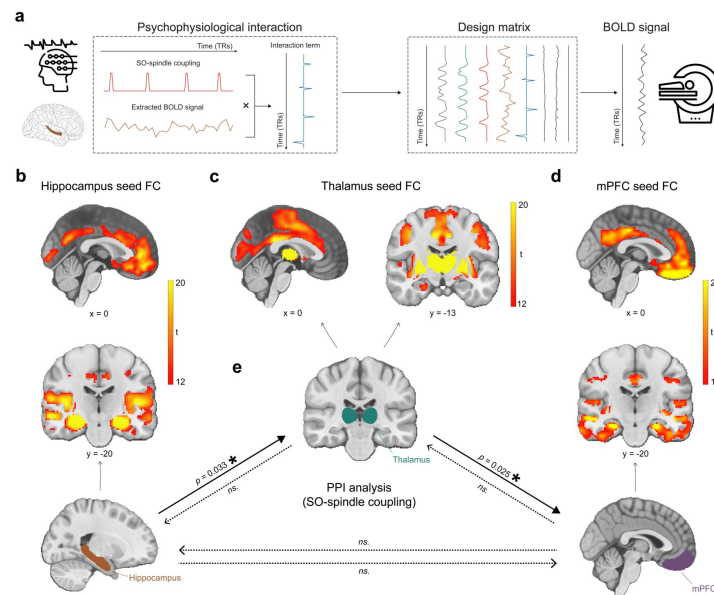


Fig. 4

Functional connectivity changes during SO-spindle coupling.

a, The PPI analysis framework for detecting brain-wide connectivity changes during SO-spindle coupling. This starts by setting a specific ROI (e.g., the hippocampus) as the seed to extract the BOLD signal (physiological condition) and using identified SO-spindle coupling events as the psychological condition to compute the interaction term. The design matrix includes the main effects of the physiological and psychological conditions, along with their interaction. This analysis examines whether whole-brain communication with the hippocampus changes as a function of SO-spindle coupling. **b**, Hippocampus-based functional connectivity with the whole brain (main effect of hippocampus BOLD signal in PPI analysis). The hippocampus ROI is bilateral, anatomically defined (bottom, orange colour). Brain-wide connectivity is shown with whole-brain FWE correction at the cluster level ($p < 1e-7$) with a cluster-forming voxel threshold of $p_{unc.} < 0.001$ for visualization purpose. **c**, Same with panel b, but based on thalamus (bilateral anatomically defined ROI). **d**, Same with panel b, but based on the mPFC (bilateral functionally defined ROI, detailed in Methods). **e**, Functional connectivity changes during SO-spindle coupling for hippocampus-based (left bottom, orange colour), thalamus-based (middle, green colour), and mPFC-based (right bottom, purple colour) connectivity. The results of ROI analysis for each direction are shown on the arrows. * $p < 0.05$, ns., not significant. Abbreviations: FC - functional connectivity, PPI - psychophysiological interaction.

Discussion

Using simultaneous EEG and fMRI during nocturnal naps in 107 subjects, we identified robust coupling between spindles and the UP-state of SOs in deep NREM sleep (N2/3 stages). This coupling was linked to distinct neural activation patterns, including significant thalamic activation during spindle occurrences, reduced activity in the DMN during SOs trough, and selective hippocampal activation when spindles and SOs coupled. Furthermore, functional connectivity analyses revealed increased connectivity from the hippocampus to the thalamus and from the thalamus to the mPFC, only during SO-spindle coupling, but not isolated SOs or spindle events alone. These results shed light on how the thalamus may coordinate hippocampal-cortical interactions during NREM sleep.

Our data provide insights into the neurobiological underpinnings of these sleep rhythms. SOs, originating mainly in neocortical areas such as the mPFC, alternate between DOWN- and UP-states. The thalamus generates sleep spindles, which in turn couple with SOs. Our finding that spindle peaks consistently occurred slightly before the UP-state peak of SOs (in 83 out of 107 participants), concurs with prior studies, including Schreiner et al. (2021) [\[1\]](#). Yet it differs from some results suggesting spindles might peak right at the SO UP-state (Hahn et al., 2020 [\[2\]](#); Helfrich et al., 2018 [\[3\]](#)). Such discrepancies could arise from differences in detection algorithms, participant age (Helfrich et al., 2018 [\[3\]](#)), or subtle variations in cortical-thalamic timing. Nonetheless, these results underscore the importance of coordinated SO-spindle interplay in supporting sleep-dependent processes.

When modeling the timing of these sleep rhythms in the fMRI, we observed hippocampal activation selectively during SO-spindle events. This suggests the possibility of triple coupling (SOs–spindles–ripples), even though our scalp EEG was not sufficiently sensitive to detect hippocampal ripples—key markers of memory replay (Buzsáki, 2015 [\[4\]](#)). Recent iEEG evidence indicates that ripples often co-occur with both spindles (Ngo, Fell, & Staresina, 2020 [\[5\]](#)) and SOs (Staresina et al., 2015 [\[6\]](#); Staresina et al., 2023 [\[7\]](#)). Therefore, the hippocampal involvement during SO-spindle events in our study may reflect memory replay from the hippocampus, propagated via thalamic spindles to distributed cortical regions.

The thalamus, known to generate spindles (Halassa et al., 2011 [\[8\]](#)), plays a key role in producing and coordinating sleep rhythms (Coulon, Budde, & Pape, 2012 [\[9\]](#); Crunelli et al., 2018 [\[10\]](#)), while the hippocampus is found essential for memory consolidation (Buzsáki, 2015 [\[4\]](#); Diba & Buzsáki, 2007 [\[11\]](#); Singh, Norman, & Schapiro, 2022 [\[12\]](#)). The increased hippocampal and thalamic activity, along with strengthened connectivity between these regions and the mPFC during SO-spindle events, underscores a hippocampal-thalamic-neocortical information flow. This aligns with recent findings suggesting the thalamus orchestrates neocortical oscillations during sleep (Schreiner et al., 2022 [\[13\]](#)). The thalamus and hippocampus thus appear central to memory consolidation during sleep, guiding information transfer to the neocortex, e.g., mPFC.

An intriguing aspect of our findings is the reduced DMN activity during SOs when modeled at the SO trough (DOWN-state). This reduced DMN activity may reflect large-scale neural inhibition characteristic of the SO trough. The DMN is typically active during internally oriented cognition (e.g., self-referential processing or mind-wandering) and is suppressed during external stimuli processing (Yeshurun, Nguyen, & Hasson, 2021 [\[14\]](#)). It is unlikely, however, that this suppression of DMN during SO events is related to a shift from internal cognition to external responses given it is during deep sleep time. Instead, it could be driven by the inherent rhythmic pattern of SOs, which makes it difficult to separate UP-from DOWN-states (the two temporal regressors were highly correlated, and similar brain activation during SOs events was obtained if modelled at the SO peak instead, **Fig. S5** [\[15\]](#)). Since the amplitude at the SO trough is consistently larger than that at the SO peak, the neural activation we detected may primarily capture the large-scale inhibition from

DOWN-state. Interestingly, no such DMN reduction was found during SO-spindle coupling, implying that coupling may involve distinct neural dynamics that partially re-engage DMN-related processes, possibly reflecting memory-related reactivation. Future research using high-temporal-resolution techniques like iEEG could clarify these possibilities.

To explore functional relevance, we employed an open-ended cognitive state decoding approach using meta-analytic data (NeuroSynth: [Yarkoni et al. \(2011\)](#)). Although this method usefully generates hypotheses about potential cognitive processes, particularly in the absence of a pre- and post-sleep memory task, it is inherently indirect. Many cognitive terms showed significant associations (16 of 50), such as “episodic memory,” “declarative memory,” and “working memory.” We focused on episodic/declarative memory given the known link with hippocampal reactivation ([Diekelmann & Born, 2010](#); [Staresina et al., 2015](#); [Staresina et al., 2023](#)). Nonetheless, these inferences regarding memory reactivation should be interpreted cautiously without direct behavioral measures. Future research incorporating explicit tasks before and after sleep would more rigorously validate these potential functional claims.

Despite providing new insights, our study has several limitations. First, our scalp EEG did not directly capture hippocampal ripples, preventing us from conclusively demonstrating triple coupling. Second, the combination of EEG-fMRI and the lack of a memory task limit our ability to parse fine-grained BOLD responses at the DOWN- vs. UP-states of SOs and link observed activations to behavioral outcomes. Third, the use of large anatomical ROIs may mask subregional contributions of specific thalamic nuclei or hippocampal subfields. Finally, without a memory task, we cannot establish a direct behavioral link between sleep-rhythm-locked activation and memory consolidation. Future studies combining techniques such as ultra-high-field fMRI or iEEG with cognitive tasks may refine our understanding of subregional network dynamics and functional significance during sleep.

In summary, we found that SO-spindle coupling during deep NREM sleep is associated with distinct activation in both the thalamus and hippocampus, accompanied by increased connectivity from the hippocampus to the thalamus and from the thalamus to the mPFC. These results emphasize the central role of the thalamus in orchestrating hippocampal-cortical interactions and highlight potential neural substrates underlying the complex interplay of SOs, spindles, and, possibly, ripples. Our open-ended decoding analyses suggest that these coordinated sleep events may be relevant for memory-related processes; however, future experiments incorporating explicit behavioral measures are necessary to confirm their functional significance. By revealing how these rhythms modulate widespread brain networks, our findings open avenues for developing targeted neuromodulatory strategies to enhance sleep-dependent cognitive functions and potentially address sleep-related disorders.

Methods

Participants & Protocol

A total of 138 healthy adults (age: 22.56 ± 0.30 ; 81 females) were recruited for the study. All participants had either normal vision or vision corrected to normal standards. None had any past psychiatric or neurological disorders. Prior to participation, they were screened for eligibility to undergo magnetic resonance imaging (MRI). The experiment received approval from Peking University ethics committee (reference number: #2015-26, #2018-11-05). Every participant provided written informed consent.

At the start of the experiment, subjects arrived early at the laboratory for registration. They signed informed consent forms, washed their hair, changed their clothes, and put on the EEG cap in preparation for the experiment. After these preliminary steps, participants entered the MRI

scanning room by around 20:30, settled down for sleep, and data collection began. Given concern for potential data drift and decline in the signal-to-noise ratio due to extended and continuous fMRI scanning, the data acquisition was segmented into several sessions for data collection. A “sleep” session was concluded either when a participant was fully awake (unable to fall asleep again or needed to use the restroom) or after 8192 seconds (the maximum scanning time of the scan, equating to 4096 volumes with 2 seconds TR).

After removing subjects with excessive head movements (mean FD > 0.5) or those whose sleep durations were under 10 minutes, the final participant count for further analyses was 107 (age: 22.40 ± 0.34 ; 63 females) with a mean scanning duration of sleep of 3.01 ± 0.14 hours.

EEG data acquisition

EEG was recorded simultaneously with fMRI data using an MR-compatible EEG amplifier system (BrainAmps MR-Plus, Brain Products, Germany), along with a specialized electrode cap. The recording was done using 64 channels in the international 10/20 system, with the reference channel positioned at FCz. In order to adhere to polysomnography (PSG) recording standards, six electrodes were removed from the EEG cap: one for electrocardiogram (ECG) recording, two for electrooculogram (EOG) recording, and three for electromyogram (EMG) recording. EEG data was recorded at a sample rate of 5000 Hz, the resistance of the reference and ground channels was kept below 10 k Ω , and the resistance of the other channels was kept below 20 k Ω . To synchronize the EEG and fMRI recordings, the BrainVision recording software (BrainProducts, Germany) was utilized to capture triggers from the MRI scanner. The EEG sampling was synchronized to the MR scanner’s 10 MHz gradient system clock, ensuring a stable gradient artifact shape over time and enabling accurate artifact removal. This was achieved via the standard clock synchronization interface of the EEG amplifier, minimizing timing jitter and drift.

MRI data acquisition

All MRI data were acquired using a 20-channel head coil on a research-dedicated 3-Tesla Siemens Magnetom Prisma MRI scanner. Earplugs and cushions were provided for noise protection and head motion restriction. We chose the 20-channel head coil because it was compatible with our EEG system and ensured sufficient signal-to-noise ratio (SNR) for both EEG and fMRI acquisition. The EEG electrode cables were guided through the lateral and posterior openings of the head coil, secured with foam padding to reduce motion and minimize MR-related artifacts. Moreover, given the extended nature of nocturnal sleep recordings, the 20-channel coil helped maintain participant comfort while still achieving high-quality simultaneous EEG-fMRI data.

For registration purposes, high-resolution T1-weighted anatomical images were acquired using a three-dimensional magnetization prepared rapid acquisition gradient-echo sequence (sequence specification: number of slices = 192, repetition time (TR) = 2,530 ms, echo time (TE) = 2.98 ms, inversion time = 1,100 ms, voxel size = $0.5 \times 0.5 \times 1$ mm³, flip angle (FA) = 7°). The participants were asked to lie quietly in the scanner during data acquisition.

Then, the “sleep” session began after the participants were instructed to try and fall asleep. For the functional scans, whole-brain images were acquired using k-space and steady-state T2*-weighted gradient echo-planar imaging (EPI) sequence that is sensitive to the BOLD contrast. This measures local magnetic changes caused by changes in blood oxygenation that accompany neural activity (sequence specification: 33 slices in interleaved ascending order, TR = 2000 ms, TE = 30 ms, voxel size = $3.5 \times 3.5 \times 4.2$ mm³, FA = 90°, matrix = 64×64 , gap = 0.7 mm). A relatively long TR and larger voxel size were chosen to optimize SNR and reduce susceptibility-related distortions, which are critical in EEG-fMRI sleep studies where head motion and physiological noise can be substantial. The longer TR allowed whole-brain coverage with sufficient temporal resolution, while the larger voxel size helped enhance BOLD sensitivity and minimize partial volume effects in deep brain structures (e.g., the thalamus and hippocampus), which are key regions of interest in this study.

EEG data preprocessing and sleep stage scoring

EEG data collected inside MRI scanner was contaminated by imaging, ballistocardiographic (BCG) and ocular artifacts. To deal with these artifacts, the preprocessing processes was divided into two parts: automated processing and manual processing.

The automated processing was carried out using custom Python scripts and the MNE toolkit (Gramfort et al., 2013 [↗](#)). This encompassed the following procedures: **I.** The raw EEG data were downsampled from 5000 Hz to 2048 Hz. This downsampled rate accelerates subsequent analyses while retaining the necessary precision to distinguish MR noise. **II.** Imaging artifacts were removed using the Average Artefact Subtraction method (Allen, Josephs, & Turner, 2000 [↗](#)). **III.** We then applied 0.1 - 30 Hz band-pass finite impulse response (FIR) filters to all channels. The EEG and EOG electrodes were re-referenced by subtracting the value from the M1 or M2 electrode (whichever was from the opposite hemisphere). The outer EMG electrodes were re-referenced to the middle one. **IV.** The signal space projection algorithm was applied to extract all epochs based on TRs and remove residual MR noise, as well as automatically extracting all heartbeat epochs using ECG electrode data for the removal of BCG noise. **V.** The data were further downsampled to 100 Hz and subjected to Independent Component Analysis (ICA).

Following this, the manual processing is carried out with labelling and removing the physiological artifacts ICs belonging to eye, muscle and residual BCG through the use of custom MATLAB scripts and the EEGLAB toolkit (Delorme & Makeig, 2004 [↗](#)). Sleep stages were then scored by YASA toolkit (Vallat & Walker, 2021 [↗](#)) after preprocessing, necessitating data from electrodes C3, left EOG, and left EMG. The classification of N2/3 stages were further verified by experts for validity.

MRI data preprocessing

Results included in this manuscript come from preprocessing performed using *fMRIPrep* 21.0.4 (Esteban et al., 2020 [↗](#); Esteban et al., 2019 [↗](#)), which is based on *Nipype* 1.6.1 (Gorgolewski et al., 2011 [↗](#)). Many internal operations of *fMRIPrep* use *Nilearn* 0.8.1 (Abraham et al., 2014 [↗](#)), mostly within the functional processing workflow. For more details of the pipeline, see <https://fmriprep.readthedocs.io/en/latest/workflows.html> [↗](#).

Conversion of data to the brain imaging data structure standard

To facilitate further analysis and sharing of data, all study data were arranged according to the Brain Imaging Data Structure (BIDS) specification using *dcm2bids* tool, which is freely available from <https://unfmontreal.github.io/Dcm2Bids/> [↗](#).

Anatomical data preprocessing

The T1-weighted (T1w) image was corrected for intensity non-uniformity (INU) with *N4BiasFieldCorrection* (Tustison et al., 2010 [↗](#)), distributed with ANTs 2.3.3 (Avants et al., 2008 [↗](#)), and used as T1w-reference throughout the workflow. The T1w-reference was then skull-stripped with a *Nipype* implementation of the *antsBrainExtraction.sh* workflow (from ANTs), using *OASIS30ANTs* as target template. Brain tissue segmentation of cerebrospinal fluid (CSF), white-matter (WM) and gray-matter (GM) was performed on the brain-extracted T1w using *fast* (FSL 6.0.5.1:57b01774, RRID:SCR_002823) (Zhang, Brady, & Smith, 2001 [↗](#)). Volume-based spatial normalization to one standard space (MNI152NLin2009cAsym) was performed through nonlinear registration with *antsRegistration* (ANTs 2.3.3), using brain-extracted versions of both T1w reference and the T1w template. The following template was selected for spatial normalization: *ICBM 152 Nonlinear Asymmetrical template version 2009c* (Fonov, 2009) [RRID:SCR_008796; TemplateFlow ID: MNI152NLin2009cAsym].

Functional data preprocessing

First, a reference volume and its skull-stripped version were generated using a custom methodology of *fMRIPrep*. Head-motion parameters with respect to the BOLD reference (transformation matrices, and six corresponding rotation and translation parameters) are estimated before any spatiotemporal filtering using *mcflirt* (Jenkinson et al., 2002) [FSL 6.0.5.1:57b01774]. BOLD runs were slice-time corrected to 0.96s (0.5 of slice acquisition range 0s–1.92s) using 3dTshift from AFNI (Cox & Hyde, 1997). The BOLD time-series (including slice-timing correction when applied) were resampled onto their original, native space by applying the transforms to correct for head-motion. These resampled BOLD time-series will be referred to as *preprocessed BOLD in original space*, or just *preprocessed BOLD*. The BOLD reference was then co-registered to the T1w reference using *mri_coreg* (FreeSurfer) followed by *flirt* (Jenkinson & Smith, 2001) [FSL 6.0.5.1:57b01774] with the boundary-based registration (Greve & Fischl, 2009) cost-function. Co-registration was configured with six degrees of freedom. Several confounding time-series were calculated based on the *preprocessed BOLD*: framewise displacement (FD), DVARS and three region-wise global signals. FD was computed using two formulations following Power (absolute sum of relative motions) (Power et al., 2014) and Jenkinson (relative root mean square displacement between affines, Jenkinson et al. (2002)). FD and DVARS are calculated for each functional run, both using their implementations in *Nipype* (following the definitions by Power et al. (2014)). The three global signals are extracted within the CSF, the WM, and the whole-brain masks. Additionally, a set of physiological regressors were extracted to allow for component-based noise correction (*CompCor*) (Behzadi et al., 2007). Principal components are estimated after high-pass filtering the *preprocessed BOLD* time-series (using a discrete cosine filter with 128s cut-off) for the two *CompCor* variants: temporal (tCompCor) and anatomical (aCompCor). tCompCor components are then calculated from the top 2% variable voxels within the brain mask. For aCompCor, three probabilistic masks (CSF, WM and combined CSF+WM) are generated in anatomical space. The implementation differs from that of Behzadi et al. (2007) in that instead of eroding the masks by 2 pixels on BOLD space, the aCompCor masks are subtracted a mask of pixels that likely contain a volume fraction of GM. This mask is obtained by thresholding the corresponding partial volume map at 0.05, and it ensures components are not extracted from voxels containing a minimal fraction of GM. Finally, these masks are resampled into BOLD space and binarized by thresholding at 0.99 (as in the original implementation). Components are also calculated separately within the WM and CSF masks. For each *CompCor* decomposition, the *k* components with the largest singular values are retained, such that the retained components' time series are sufficient to explain 50 percent of variance across the nuisance mask (CSF, WM, combined, or temporal). The remaining components are dropped from consideration. Frames that exceeded a threshold of 0.5 mm FD or 1.5 standardised DVARS were annotated as motion outliers.

The BOLD time-series were resampled into standard space, generating a *preprocessed BOLD run in MNI152NLin2009cAsym space*. First, a reference volume and its skull-stripped version were generated using a custom methodology of *fMRIPrep*. All resamplings can be performed with a *single interpolation step* by composing all the pertinent transformations (i.e., head-motion transform matrices, susceptibility distortion correction when available, and co-registrations to anatomical and output spaces). Gridded (volumetric) resamplings were performed using *antsApplyTransforms* (ANTs), configured with Lanczos interpolation to minimize the smoothing effects of other kernels (Lanczos, 1964). Non-gridded (surface) resamplings were performed using *mri_vol2surf* (FreeSurfer).

EEG sleep rhythms analysis

In line with other studies examining the coordination between SOs and spindles, particularly their spatial distribution (Massimini et al., 2004; Molle et al., 2011), EEG data from the F3 electrode were used to quantify the occurrence of SOs and spindles. Using the established detection algorithms (detailed below), we identified SOs and sleep spindles for each subject.

Detection of SOs

Data were first bandpass-filtered between 0.16 and 1.25 Hz (Butterworth filter, order 3, bidirectional filtering for zero phase). After identifying all positive-to-negative zero crossings, potential SOs were defined based on the interval between consecutive zero crossings (ranging from 0.8 s to 3 s). For each potential SO, we calculated the amplitude range (peak minus trough). Only those exceeding the 75th percentile in amplitude were labeled as SOs, following previous work (Schreiner et al., 2021 [DOI](#)).

Detection of sleep spindles

Data were bandpass-filtered between 12 and 16 Hz (Butterworth filter, order 3, bidirectional filtering for zero phase). The root mean square (RMS) of the filtered signal was computed with a 200 ms sliding time window. The amplitude threshold for spindles was set at the 75th percentile of RMS values. Any RMS values that surpassed this threshold for 0.5 s to 3 s were identified as spindles (Staresina et al., 2015 [DOI](#)).

Detection of SO-spindle couplings

From the detected SOs and spindles, we identified the peak time of each spindle. Within each SO interval, we checked for a spindle peak; if found, it was recorded as an SO-spindle coupling event. For every SO-spindle coupling event, an epoch was created time-locked to the SO trough (as the central reference), following Schreiner et al. (2021) [DOI](#). We extracted data in a [-4 s to 4 s] window around this point, forming the epoch for each coupling event.

We note that the above methods for detecting SOs, spindles, and their couplings were originally developed for N2 and N3 sleep data, based on the specific characteristics of these stages. These methods are widely recognized in sleep research (Hahn et al., 2020 [DOI](#); Helfrich et al., 2019 [DOI](#); Helfrich et al., 2018 [DOI](#); Ngo, Fell, & Staresina, 2020 [DOI](#); Schreiner et al., 2022 [DOI](#); Schreiner et al., 2021 [DOI](#); Staresina et al., 2015 [DOI](#); Staresina et al., 2023 [DOI](#)). However, because this percentile-based detection approach will inherently identify a certain number of events if applied to other stages (e.g., N1 and REM), the nature of these events in those stages remains unclear compared to N2/N3. We nevertheless identified and reported the detailed descriptive statistics of these sleep rhythms in all sleep stages, under the same operational definitions, both for completeness and as a sanity check. Within the same subject, there should be more SOs, spindles, and their couplings in N2/N3 than in N1 or REM (see also **Figure S2-S4** [DOI](#), **Table S1-S4** [DOI](#)).

Event-related potentials (ERP) analysis

After completing the detection of each sleep rhythm event, we performed ERP analyses for SOs, spindles, and coupling events in different sleep stages. Specifically, for SO events, we took the trough of the DOWN-state of each SO as the zero-time point, then extracted data in a [-2 s to 2 s] window from the broadband (0.1–30 Hz) EEG and used [-2 s to -0.5 s] for baseline correction; the results were then averaged across 107 subjects (see **Fig. S2a** [DOI](#)). For spindle events, we used the peak of each spindle as the zero-time point and applied the same data extraction window and baseline correction before averaging across 107 subjects (see **Fig. S2b** [DOI](#)). Finally, for SO-spindle coupling events, we followed the same procedure used for SO events (see **Fig. 2a** [DOI](#), **Figs. S3-S4** [DOI](#)).

Time-frequency analysis

We performed this analysis on the SO-spindle coupling epochs using the FieldTrip toolbox (Oostenveld et al., 2011 [DOI](#)). Fourier analysis was executed on the data, utilizing a sliding time window with a forward shift of 50 ms. The window length was defined as five cycles of the present

frequency (frequency range: 1-30 Hz, step size: 1 Hz). Before Fourier analysis, windowed data segments underwent multiplication by a Hanning taper. Subsequently, power values were normalized to z-scores within the time range of [-4 s to 4 s], selecting a longer duration to prevent edge artifacts from low-frequency activity in the target time window of [-1.5 s to 1.5 s].

Preferred phase analysis

All SO-spindle coupling epochs were filtered to the SO frequency band (0.16-1.25 Hz, 3rd-order Butterworth filter), and the instantaneous SO phase was determined through the Hilbert transform. Using the saved spindle peak times for each SO-spindle coupling, the preferred phase was deduced. The distribution of these phases across participants was then tested for uniformity using the Rayleigh test.

fMRI GLM analysis

All fMRI GLM analyses were conducted using only data from the N2/3 stages of NREM sleep. Segments of sleep data from these stages were concatenated for each participant. After pre-processing, the fMRI data was smoothed with an 8 mm FWHM kernel. All images underwent a temporal high-pass filter (width of 100 s), and the autocorrelation of the hemodynamic response was modelled via an AR(1) model. We then conducted General Linear Model (GLM) analyses to identify whole-brain activations associated with sleep rhythms and their coupling. We incorporated nuisance regressors estimated during the preprocessing with fMRIPrep: six rigid-body motion-correction parameters identified during realignment (comprising three translation and rotation parameters), the mean White Matter, and the mean Cerebral Spinal Fluid signals. The cosine drift model was employed to detect and eliminate data fluctuations, thereby enhancing the accuracy of actual experimental effects. The GLMs were designed to evaluate the influence of experimental conditions on BOLD responses. All whole-brain analyses were thresholded using whole-brain FWE corrected at the cluster level ($p < 0.05$), with a cluster-forming voxel threshold of $p_{\text{unc.}} < 0.001$, unless otherwise specified.

The fMRI GLM analysis was designed to look for brain activations pertinent to SO, spindle, and SO-spindle coupling. Drawing from the EEG results concerning the timing of SO and spindle, we identified the trough of the SO as the onset for SO events and the peak of the spindle as the onset for spindle events. We chose the trough of the SO and the peak of the spindle as the onset points, primarily due to the larger amplitudes observed at these moments (thereby more robust to noise). This was also the common choice in previous EEG studies on SO and spindles (Ngo, Fell, & Staresina, 2020 [DOI](#); Staresina et al., 2015 [DOI](#); Staresina et al., 2023 [DOI](#)). To model SO-spindle coupling in fMRI, we followed Schreiner et al. (2021) [DOI](#), and defining SO-spindle complex as the peak of spindle appeared within 1.5 seconds after SO trough. When such a peak was found, that particular SO was labelled as an SO-spindle event, with its SO trough labelled as the event onset. After convolving these events with the canonical Hemodynamic Response Function (HRF) of SPM and downsampling them to align with the fMRI time resolution, they were integrated as regressors in the design matrix for GLM analysis, allowing us to determine the effects of SO and spindle, as well as the SO-spindle coupling (Fig. 3b-d [DOI](#)).

We have also conducted a control analysis aimed to distinguish brain activations associated with DOWN- vs. UP state of SOs. Drawing again from EEG data on SO and spindle, we randomly selected half of the SO events using the peak as the onset and the other half using the trough, to reduce collinearity between the two regressors. As with the main analysis, we identified the spindle peak as the onset for spindle events (Fig. S5 [DOI](#)).

ROI analysis

The objective of our ROI analysis was to test the activation of specific brain regions during distinct sleep rhythms. Within each ROI, the beta values at the subject-level were averaged across all voxels to facilitate subsequent statistical inferences (one-sample t test). We defined the mDMN and mPFC ROIs functionally based on the meta-analysis results in Neurosynth (Yarkoni et al., 2011 [↗](#)), by the keyword “medial default mode network” and “ventromedial prefrontal cortex” respectively. In addition to mDMN and mPFC, we defined other ROIs anatomically, using the high-resolution Harvard-Oxford Atlas probabilistic atlas (Desikan et al., 2006 [↗](#)).

Open-ended Cognitive State Functional Decoding

For functional decoding, we followed the approach in Margulies et al. (2016) [↗](#). We used the NeuroSynth (Yarkoni et al., 2011 [↗](#)) meta-analytic database (www.neurosynth.org [↗](#)) to assess topic terms associated with the sleep rhythms activation patterns. ROI masks were created from the whole-brain activation patterns associated with the three sleep rhythms in **Fig. 3b-d** [↗](#). These activation patterns were binarized using a threshold of $p < 0.001$, forming a brain mask for each sleep rhythm.

For each ROI mask, the analysis output was a z -statistic associated with a feature term, and the terms were ranked based on the weighted sum of their z -scores. The feature terms were derived from the most general set of 50 topic terms available in the NeuroSynth v3 dataset (https://github.com/NeuroanatomyAndConnectivity/gradient_analysis/blob/master/gradient_data/neurosynth/v3-topics-50-keys.txt [↗](#)).

From the 50 topics, 16 exceeded the threshold of $z > 1.96$ ($p < 0.05$). Two terms were removed as they did not represent coherent cognitive functions, leaving 14 relevant topic terms (**Fig. S7** [↗](#)). These included: episodic memory, declarative memory, working memory, task representation, language, learning, faces, visuospatial processing, category recognition, cognitive control, reading, cued attention, inhibition, and action.

PPI analysis

PPI (psychophysiological interaction) analysis is designed to quantify context-dependent alterations in effective connectivity between brain regions. For instance, it can determine whether hippocampus-based functional connectivity with the thalamus varies based on the occurrence of an SO-spindle coupling event. We conducted nine whole-brain PPI analyses with *nilearn* to investigate interactions based in the hippocampus, thalamus or mPFC during SO-spindle coupling, isolated SO and isolated spindle, utilizing the same ROIs. Both the main effects of physiological and physiological condition, as well as their interaction were modelled in the design matrix for PPI analysis. The main effect of physiological term produced functional connectivity maps of the entire brain in relation to the specified ROI (**Fig. 4b-d** [↗](#)). While the interaction term asked for the *change* of functional connectivity along with the SO-spindle coupling (**Fig. 4e** [↗](#)).

Data and materials availability

The data that support the findings can be provided by the corresponding author Y.L. pending scientific review and a completed material transfer agreement. Requests for the data should be submitted to: <https://doi.org/10.5061/dryad.2fqz612x0> [↗](#). The analysis code will be publicly available on https://gitlab.com/liu_lab/eeg-fmri-sleep [↗](#) upon publication.

Supplementary figures and tables

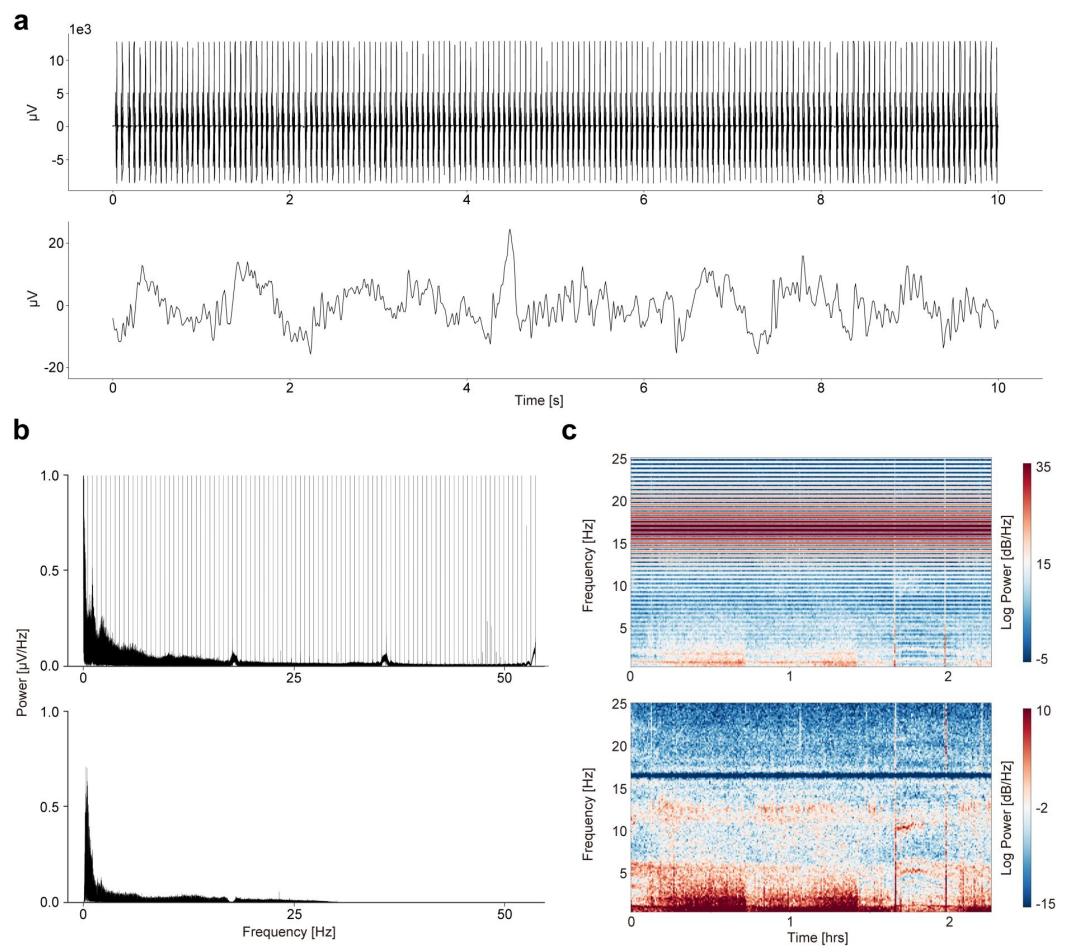


Fig. S1

Removal of MRI gradient noise from simultaneous collected EEG data.

a, Time series of both raw and preprocessed EEG data. The top row depicts the raw EEG data, which contains noise primarily from the MRI gradient magnetic field and electrocardiographic artifacts. The bottom row showcases the preprocessed EEG data (detailed in Methods). **b**, Power spectral density of the raw and preprocessed EEG data estimated by the fast Fourier transform. The raw EEG data is shown in the top row, while the preprocessed EEG data is in the bottom row. **c**, Time-frequency spectrogram of the raw and preprocessed EEG data, by the short-time Fourier transform. The top row represents the raw EEG data, and the bottom row displays the preprocessed EEG data.

Fig. S2

ERPs of SOs and spindles coupling during different sleep stages across all 107 subjects.

a. ERP of SOs in different sleep stages using the broadband (0.1–30 Hz) EEG data. We align the trough of the DOWN-state of each SO at time zero (see Methods for details). The orange line represents the SO ERP in the N1 stage, the black line represents the SO ERP in the N2&N3 stage, and the green line represents the SO ERP in the REM stage. **b.** ERP of spindles in different sleep stages using the broadband (0.1–30 Hz) EEG data. We align the peak of each spindle at time zero (see Methods for details). The color scheme is the same as in panel a.

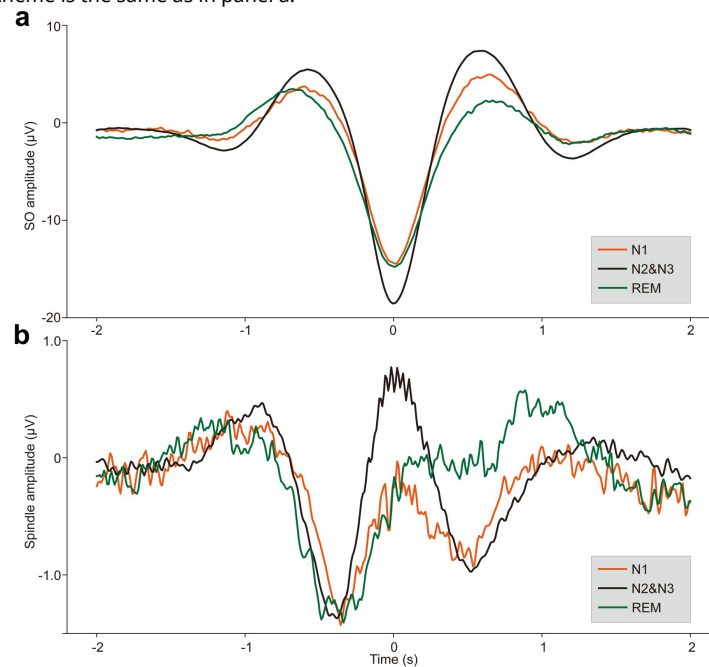


Fig. S3

ERP and time-frequency patterns of SO-spindle coupling in the N1 stage.

The averaged temporal frequency pattern and ERP across all instances of SO-spindle coupling, computed over all subjects, following the same procedure as in [Fig. 2a](#), but for N1 stage.

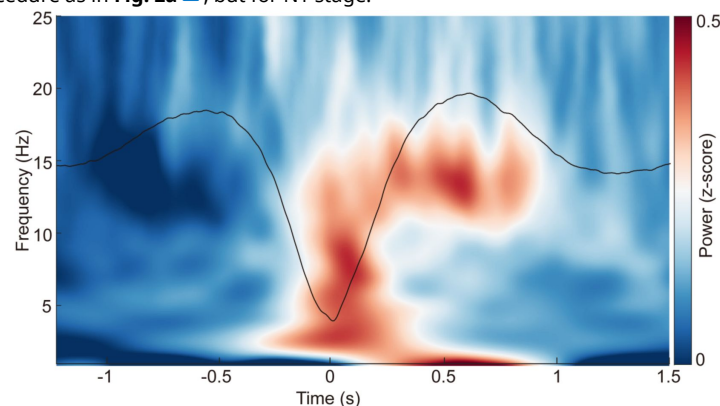


Fig. S4

ERP and time-frequency patterns of SO-spindle coupling in the REM stage.

The averaged temporal frequency pattern and ERP across all instances of SO-spindle coupling, computed over all subjects, again following the same procedure as in [Fig. 2a](#), but for REM stage.

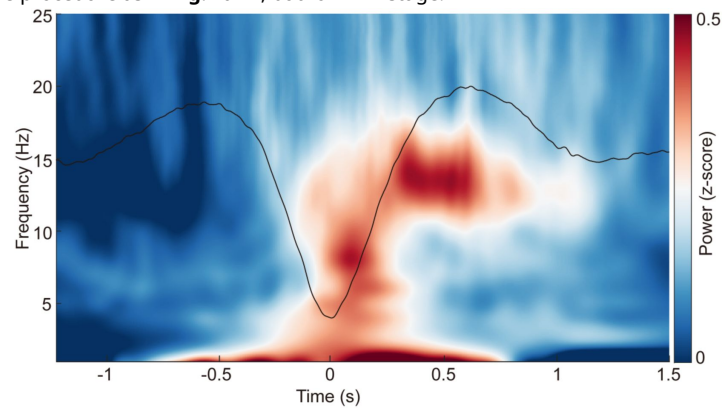


Fig. S5

Brain-wide activity between SO UP-state (peak) and DOWN-state (trough).

a, Brain activity with SO DOWN-state (trough) modelled as event onset, whole-brain FWE corrected at the cluster level ($p < 0.05$) with a cluster-forming voxel threshold of $p_{unc} < 0.001$. **b**, Brain activity with SO UP-state (peak) modelled as event onset. **c**, Differences in brain activity corresponding to SO UP-state and SO DOWN-state. The whole-brain results were displayed at an uncorrected threshold of $p < 0.01$ for visualization purpose only. No brain region was found significant in this contrast.

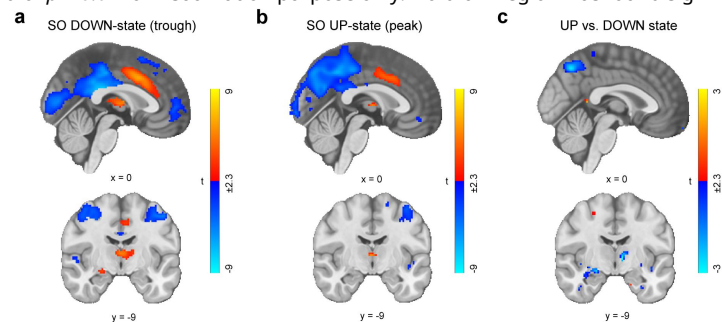


Fig. S6

Influence of the percentile threshold for SO detection on hippocampal activation (ROI) during SO-spindle coupling.

We changed the percentile threshold for SO event detection in the EEG data analysis and then reconstructed the GLM design matrix based on the SO events detected at each threshold. The brain-wide activation pattern of SO-spindle couplings in the N2/3 stage was extracted using the same method as shown in [Fig. 3](#). The gray horizontal line represents the significant range (71%–80%). * $p < 0.05$.

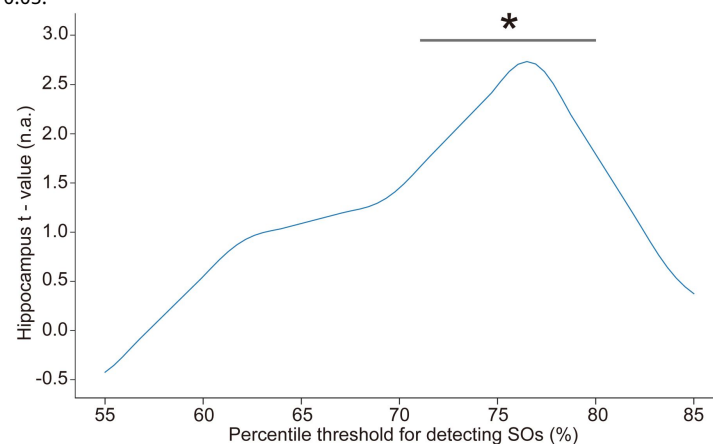
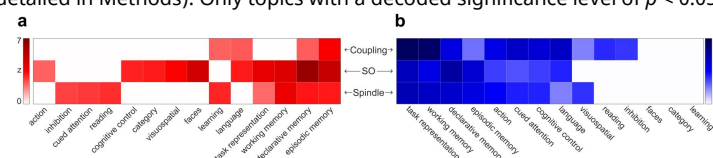


Fig. S7

Functional decoding using the ROI association method in Neurosynth.

a, Decoding results using positive activation. **b**, Decoding results using negative activation. Each row corresponds to the brain-wide activation patterns for sleep rhythms shown in [Fig. 3b-d](#), while each column corresponds to topics in the Neurosynth database (detailed in Methods). Only topics with a decoded significance level of $p < 0.05$ are displayed.



Descriptive statistics of demographic and sleep	Mean ± SE
	(n = 107)
Age	22.40 ± 0.34
Total Recording time (min)	180.60 ± 8.40
Wake time (min)	84.60 ± 5.00
Total Sleep time (min)	95.98 ± 5.67
N1 time (min)	13.99 ± 1.19
N2 time (min)	56.21 ± 4.05
N3 time (min)	16.53 ± 1.35
REM time (min)	9.27 ± 0.86
N1 ratio (%)	14.57 ± 1.06
N2 ratio (%)	58.55 ± 1.70
N3 ratio (%)	17.22 ± 1.21
REM ratio (%)	9.66 ± 0.90
Sleep Onset Latency (min)	9.75 ± 0.85
N2 latency (min)	14.58 ± 1.15
N3 latency (min)	33.65 ± 3.02
Sleep Efficiency (%)	52.96 ± 1.60

Table S1

Descriptive results of demographic information and sleep characteristics.

Note: The total recorded time is equal to the awake time plus the total sleep time. The sleep onset latency is the time taken to reach the first sleep epoch. The Sleep Efficiency is the ratio of actual sleep time to total recording time.

Table S2

Statistics of sleep duration, SO, spindle and coupling event numbers and densities for all 107 subjects during N1 stage.

Subject ID	N1 Sleep Stage						
	Time (min)	Number of SO	Number of spindle	Number of coupling	Density of SO (times/min)	Density of spindle (times/min)	Density of coupling (times/min)
1	8.38	60	31	14	7.16	3.70	1.67
2	5.21	49	22	14	9.40	4.22	2.69
3	3.24	28	15	9	8.63	4.62	2.77
4	16.61	197	60	55	11.86	3.61	3.31
5	3.38	40	10	5	11.85	2.96	1.48
6	9.74	104	35	28	10.67	3.59	2.87
7	8.84	46	32	20	5.20	3.62	2.26
8	19.51	147	38	13	7.53	1.95	0.67
9	2.28	17	2	0	7.47	0.88	0.00
10	25.71	234	95	57	9.10	3.70	2.22
11	8.04	35	18	7	4.35	2.24	0.87
12	4.54	10	6	0	2.20	1.32	0.00
13	4.91	36	23	15	7.33	4.68	3.05
14	3.41	22	14	9	6.45	4.11	2.64
15	4.91	52	8	5	10.59	1.63	1.02
16	21.98	58	34	17	2.64	1.55	0.77
17	1.31	0	0	0	0.00	0.00	0.00
18	0.91	0	0	0	0.00	0.00	0.00
19	2.54	11	4	4	4.33	1.57	1.57
20	23.71	266	100	54	11.22	4.22	2.28
21	35.81	322	151	109	8.99	4.22	3.04
22	25.58	146	61	41	5.71	2.38	1.60
23	3.41	21	5	3	6.16	1.47	0.88
24	20.68	224	73	59	10.83	3.53	2.85
25	47.74	543	224	124	11.37	4.69	2.60
26	20.91	130	63	42	6.22	3.01	2.01
27	7.91	92	46	21	11.63	5.82	2.65
28	10.71	94	49	38	8.78	4.58	3.55
29	22.71	229	100	63	10.08	4.40	2.77
30	2.84	28	12	7	9.85	4.22	2.46
31	7.41	60	20	17	8.10	2.70	2.29
32	0.98	6	2	1	6.14	2.05	1.02
33	3.21	38	18	12	11.84	5.61	3.74
34	7.81	67	29	16	8.58	3.71	2.05
35	4.44	31	13	8	6.98	2.93	1.80
36	41.34	423	145	107	10.23	3.51	2.59
37	5.31	35	18	8	6.59	3.39	1.51
38	13.34	140	53	46	10.49	3.97	3.45
39	20.34	217	61	40	10.67	3.00	1.97
40	5.74	51	19	14	8.88	3.31	2.44
41	8.11	113	24	20	13.93	2.96	2.47
42	25.68	271	108	70	10.55	4.21	2.73
43	23.21	204	58	35	8.79	2.50	1.51
44	16.94	195	69	39	11.51	4.07	2.30
45	6.31	58	32	18	9.19	5.07	2.85
46	7.24	80	23	13	11.04	3.18	1.79
47	9.14	78	36	24	8.53	3.94	2.62
48	35.71	361	187	106	10.11	5.24	2.97
49	7.01	71	23	19	10.13	3.28	2.71
50	27.18	174	71	52	6.40	2.61	1.91
51	13.38	129	50	34	9.64	3.74	2.54
52	17.54	71	28	5	4.05	1.60	0.29
53	24.31	269	104	83	11.07	4.28	3.41
54	11.54	131	60	42	11.35	5.20	3.64
55	10.14	93	27	14	9.17	2.66	1.38
56	4.41	49	21	16	11.11	4.76	3.63

Table S2 (continued)

57	5.01	34	21	11	6.79	4.19	2.20
58	6.01	56	28	13	9.32	4.66	2.16
59	37.38	404	139	103	10.81	3.72	2.76
60	7.08	68	27	13	9.61	3.82	1.84
61	8.74	86	43	23	9.84	4.92	2.63
62	17.61	181	56	48	10.28	3.18	2.73
63	48.61	388	187	117	7.98	3.85	2.41
64	9.74	51	16	9	5.23	1.64	0.92
65	22.21	215	96	80	9.68	4.32	3.60
66	36.78	312	149	98	8.48	4.05	2.66
67	33.51	263	110	67	7.85	3.28	2.00
68	10.71	112	36	19	10.46	3.36	1.77
69	58.34	608	213	161	10.42	3.65	2.76
70	14.21	97	51	31	6.83	3.59	2.18
71	21.98	170	85	61	7.74	3.87	2.78
72	55.31	600	203	120	10.85	3.67	2.17
73	24.61	207	89	81	8.41	3.62	3.29
74	23.31	252	96	81	10.81	4.12	3.47
75	4.74	34	17	9	7.17	3.58	1.90
76	37.11	360	145	97	9.70	3.91	2.61
77	14.41	146	64	53	10.13	4.44	3.68
78	12.21	134	54	33	10.97	4.42	2.70
79	6.18	66	14	8	10.69	2.27	1.30
80	8.18	76	27	15	9.29	3.30	1.83
81	2.84	29	8	3	10.20	2.81	1.06
82	7.41	75	29	21	10.12	3.91	2.83
83	1.91	18	7	3	9.42	3.66	1.57
84	2.14	21	7	3	9.80	3.27	1.40
85	10.38	47	37	12	4.53	3.57	1.16
86	10.34	89	53	28	8.60	5.12	2.71
87	5.98	60	19	11	10.04	3.18	1.84
88	0.41	0	0	0	0.00	0.00	0.00
89	10.91	113	46	29	10.36	4.22	2.66
90	9.54	100	37	28	10.48	3.88	2.93
91	4.18	49	18	12	11.73	4.31	2.87
92	2.14	19	5	2	8.86	2.33	0.93
93	10.11	116	38	38	11.47	3.76	3.76
94	1.91	3	0	0	1.57	0.00	0.00
95	8.68	76	39	22	8.76	4.49	2.54
96	30.71	344	123	87	11.20	4.01	2.83
97	11.71	135	40	36	11.53	3.42	3.07
98	11.91	108	55	37	9.07	4.62	3.11
99	16.41	133	84	51	8.10	5.12	3.11
100	7.34	80	26	15	10.89	3.54	2.04
101	6.41	39	24	11	6.08	3.74	1.72
102	9.24	100	45	31	10.82	4.87	3.35
103	3.34	15	8	3	4.49	2.39	0.90
104	13.41	142	64	45	10.59	4.77	3.36
105	10.58	105	44	39	9.93	4.16	3.69
106	6.48	63	25	25	9.73	3.86	3.86
107	21.51	217	75	50	10.09	3.49	2.32
Mean	13.99	129.64	51.89	34.44	8.70	3.48	2.23
SE	1.19	12.18	4.78	3.27	0.26	0.12	0.09

Table S3

Statistics of sleep duration, SO, spindle and coupling event numbers and densities for all 107 subjects during N2&N3 stage.

Subject ID	N2&N3 Sleep Stage						
	Time (min)	Number of SO	Number of spindle	Number of coupling	Density of SO (times/min)	Density of spindle (times/min)	Density of coupling (times/min)
1	89.00	820	401	242	9.21	4.51	2.72
2	125.00	1417	659	298	11.34	5.27	2.38
3	22.50	225	115	75	10.00	5.11	3.33
4	87.00	1059	326	175	12.17	3.75	2.01
5	44.50	425	190	115	9.55	4.27	2.58
6	21.00	222	78	57	10.57	3.71	2.71
7	42.50	185	160	91	4.35	3.76	2.14
8	167.50	1865	882	502	11.13	5.27	3.00
9	25.50	112	28	24	4.39	1.10	0.94
10	65.00	715	322	197	11.00	4.95	3.03
11	69.00	511	302	155	7.41	4.38	2.25
12	62.50	413	316	162	6.61	5.06	2.59
13	89.00	826	449	283	9.28	5.04	3.18
14	41.00	390	191	93	9.51	4.66	2.27
15	105.00	359	96	51	3.42	0.91	0.49
16	184.50	708	425	228	3.84	2.30	1.24
17	97.50	395	306	155	4.05	3.14	1.59
18	23.50	212	86	75	9.02	3.66	3.19
19	17.50	137	48	39	7.83	2.74	2.23
20	102.00	1196	462	290	11.73	4.53	2.84
21	139.00	1236	671	359	8.89	4.83	2.58
22	56.50	684	284	166	12.11	5.03	2.94
23	43.00	508	182	117	11.81	4.23	2.72
24	75.50	872	313	194	11.55	4.15	2.57
25	136.00	1616	792	408	11.88	5.82	3.00
26	29.00	190	118	63	6.55	4.07	2.17
27	73.50	849	408	229	11.55	5.55	3.12
28	249.50	2355	1170	806	9.44	4.69	3.23
29	54.00	603	224	137	11.17	4.15	2.54
30	7.50	34	16	13	4.53	2.13	1.73
31	30.50	269	139	91	8.82	4.56	2.98
32	53.00	550	258	128	10.38	4.87	2.42
33	24.00	290	128	68	12.08	5.33	2.83
34	44.00	401	182	111	9.11	4.14	2.52
35	14.00	120	62	36	8.57	4.43	2.57
36	177.50	1925	782	518	10.85	4.41	2.92
37	113.50	1290	569	345	11.37	5.01	3.04
38	113.50	1286	570	303	11.33	5.02	2.67
39	72.50	660	230	134	9.10	3.17	1.85
40	119.00	827	284	176	6.95	2.39	1.48
41	43.00	438	176	101	10.19	4.09	2.35
42	79.50	917	324	154	11.53	4.08	1.94
43	197.00	1166	559	369	5.92	2.84	1.87
44	93.00	1136	362	205	12.22	3.89	2.20
45	159.00	696	411	227	4.38	2.58	1.43
46	53.00	575	165	102	10.85	3.11	1.92
47	143.00	596	317	207	4.17	2.22	1.45
48	149.00	1514	850	515	10.16	5.70	3.46
49	27.00	312	102	61	11.56	3.78	2.26
50	110.00	707	339	238	6.43	3.08	2.16
51	42.50	439	144	71	10.33	3.39	1.67
52	52.00	275	171	74	5.29	3.29	1.42
53	34.00	343	136	91	10.09	4.00	2.68
54	69.00	820	359	198	11.88	5.20	2.87
55	81.00	940	327	197	11.60	4.04	2.43
56	39.50	448	209	124	11.34	5.29	3.14

Table S3 (continued)

57	32.50	301	140	76	9.26	4.31	2.34
58	51.50	525	218	119	10.19	4.23	2.31
59	55.50	561	212	150	10.11	3.82	2.70
60	114.00	1188	485	279	10.42	4.25	2.45
61	59.50	713	373	194	11.98	6.27	3.26
62	70.50	795	250	204	11.28	3.55	2.89
63	105.00	693	432	231	6.60	4.11	2.20
64	57.00	653	347	172	11.46	6.09	3.02
65	60.50	577	268	151	9.54	4.43	2.50
66	120.50	1203	622	390	9.98	5.16	3.24
67	199.00	915	521	274	4.60	2.62	1.38
68	50.00	595	183	118	11.90	3.66	2.36
69	127.50	1458	592	366	11.44	4.64	2.87
70	142.50	323	89	46	2.27	0.62	0.32
71	123.50	1261	460	324	10.21	3.72	2.62
72	65.00	195	80	55	3.00	1.23	0.85
73	92.00	1035	413	238	11.25	4.49	2.59
74	128.00	1500	638	307	11.72	4.98	2.40
75	31.50	289	153	81	9.17	4.86	2.57
76	149.00	1302	631	331	8.74	4.23	2.22
77	57.00	605	271	166	10.61	4.75	2.91
78	94.50	1097	448	247	11.61	4.74	2.61
79	58.50	704	259	126	12.03	4.43	2.15
80	131.00	709	625	236	5.41	4.77	1.80
81	42.50	489	183	126	11.51	4.31	2.96
82	25.50	284	124	89	11.14	4.86	3.49
83	32.50	383	166	110	11.78	5.11	3.38
84	48.00	488	173	117	10.17	3.60	2.44
85	48.00	376	194	105	7.83	4.04	2.19
86	56.00	499	300	154	8.91	5.36	2.75
87	49.00	583	214	106	11.90	4.37	2.16
88	35.50	445	143	73	12.54	4.03	2.06
89	44.00	454	207	121	10.32	4.70	2.75
90	24.00	258	115	59	10.75	4.79	2.46
91	30.00	369	149	83	12.30	4.97	2.77
92	49.00	592	176	93	12.08	3.59	1.90
93	7.00	82	30	26	11.71	4.29	3.71
94	101.50	1235	503	324	12.17	4.96	3.19
95	26.00	322	134	76	12.38	5.15	2.92
96	21.00	239	87	71	11.38	4.14	3.38
97	40.50	481	177	108	11.88	4.37	2.67
98	90.50	886	410	210	9.79	4.53	2.32
99	29.00	243	138	81	8.38	4.76	2.79
100	26.00	309	96	48	11.88	3.69	1.85
101	79.00	609	310	173	7.71	3.92	2.19
102	21.50	234	117	58	10.88	5.44	2.70
103	71.00	871	309	165	12.27	4.35	2.32
104	41.50	428	202	130	10.31	4.87	3.13
105	30.50	390	131	81	12.79	4.30	2.66
106	52.50	590	278	160	11.24	5.30	3.05
107	8.50	72	31	19	8.47	3.65	2.24
Mean	72.74	668.10	298.90	172.80	9.64	4.19	2.46
SE	4.68	42.64	20.40	12.18	0.25	0.10	0.06

Table S4

Statistics of sleep duration, SO, spindle and coupling event numbers and densities for all 107 subjects during REM stage.

Subject ID	REM Sleep Stage						
	Time (min)	Number of SO	Number of spindle	Number of coupling	Density of SO (times/min)	Density of spindle (times/min)	Density of coupling (times/min)
1	2.00	13	7	6	6.50	3.50	3.00
2	4.00	26	12	9	6.50	3.00	2.25
3	4.50	39	17	12	8.67	3.78	2.67
4	7.00	48	14	11	6.86	2.00	1.57
5	3.50	35	13	7	10.00	3.71	2.00
6	24.50	264	48	40	10.78	1.96	1.63
7	5.00	18	21	3	3.60	4.20	0.60
8	21.00	215	78	61	10.24	3.71	2.90
9	4.50	27	20	13	6.00	4.44	2.89
10	15.00	153	57	51	10.20	3.80	3.40
11	4.50	33	18	11	7.33	4.00	2.44
12	4.50	22	10	6	4.89	2.22	1.33
13	2.00	10	7	7	5.00	3.50	3.50
14	0.50	0	0	0	0.00	0.00	0.00
15	16.00	186	39	30	11.63	2.44	1.88
16	11.50	51	26	23	4.43	2.26	2.00
17	12.50	51	30	14	4.08	2.40	1.12
18	4.00	35	9	9	8.75	2.25	2.25
19	11.50	108	40	38	9.39	3.48	3.30
20	5.50	55	20	14	10.00	3.64	2.55
21	12.00	105	59	34	8.75	4.92	2.83
22	0.50	0	0	0	0.00	0.00	0.00
23	11.50	73	42	22	6.35	3.65	1.91
24	8.00	81	27	18	10.13	3.38	2.25
25	32.50	381	120	79	11.72	3.69	2.43
26	2.50	10	9	6	4.00	3.60	2.40
27	7.50	94	32	26	12.53	4.27	3.47
28	5.50	42	17	14	7.64	3.09	2.55
29	17.50	178	71	43	10.17	4.06	2.46
30	6.00	60	24	23	10.00	4.00	3.83
31	13.00	85	37	33	6.54	2.85	2.54
32	0.50	0	0	0	0.00	0.00	0.00
33	1.00	9	5	2	9.00	5.00	2.00
34	9.50	85	41	27	8.95	4.32	2.84
35	2.50	16	9	8	6.40	3.60	3.20
36	9.50	83	32	26	8.74	3.37	2.74
37	24.00	260	137	69	10.83	5.71	2.88
38	4.00	38	17	16	9.50	4.25	4.00
39	13.00	137	49	30	10.54	3.77	2.31
40	8.50	81	28	23	9.53	3.29	2.71
41	7.00	74	20	13	10.57	2.86	1.86
42	17.00	178	82	63	10.47	4.82	3.71
43	25.00	165	93	68	6.60	3.72	2.72
44	15.50	170	68	43	10.97	4.39	2.77
45	7.50	67	40	21	8.93	5.33	2.80
46	0.50	0	0	0	0.00	0.00	0.00
47	1.00	7	5	3	7.00	5.00	3.00
48	46.00	428	286	99	9.30	6.22	2.15
49	0.50	0	0	0	0.00	0.00	0.00
50	29.50	223	104	64	7.56	3.53	2.17
51	19.50	204	59	46	10.46	3.03	2.36
52	8.00	18	10	9	2.25	1.25	1.13
53	6.00	62	21	14	10.33	3.50	2.33
54	6.00	42	21	11	7.00	3.50	1.83
55	1.00	4	2	1	4.00	2.00	1.00
56	20.00	243	73	51	12.15	3.65	2.55

Table S4 (continued)

57	0.50	0	0	0	0.00	0.00	0.00
58	6.00	51	22	17	8.50	3.67	2.83
59	45.00	501	212	91	11.13	4.71	2.02
60	0.50	0	0	0	0.00	0.00	0.00
61	8.00	72	42	26	9.00	5.25	3.25
62	10.50	116	34	34	11.05	3.24	3.24
63	5.50	40	24	14	7.27	4.36	2.55
64	7.50	78	49	26	10.40	6.53	3.47
65	9.00	87	61	24	9.67	6.78	2.67
66	22.00	200	98	50	9.09	4.45	2.27
67	22.00	199	88	54	9.05	4.00	2.45
68	18.50	206	62	46	11.14	3.35	2.49
69	5.50	43	13	11	7.82	2.36	2.00
70	10.00	88	22	21	8.80	2.20	2.10
71	29.50	255	112	79	8.64	3.80	2.68
72	0.00	0	0	0	/	/	/
73	14.50	108	53	38	7.45	3.66	2.62
74	13.50	139	57	37	10.30	4.22	2.74
75	3.50	25	13	8	7.14	3.71	2.29
76	4.50	35	16	12	7.78	3.56	2.67
77	14.00	139	70	43	9.93	5.00	3.07
78	26.00	297	150	72	11.42	5.77	2.77
79	2.50	19	7	6	7.60	2.80	2.40
80	6.50	27	21	11	4.15	3.23	1.69
81	4.00	35	14	8	8.75	3.50	2.00
82	3.00	30	11	9	10.00	3.67	3.00
83	10.50	113	43	31	10.76	4.10	2.95
84	1.00	6	1	1	6.00	1.00	1.00
85	8.00	88	33	20	11.00	4.13	2.50
86	0.00	0	0	0	/	/	/
87	3.50	31	11	11	8.86	3.14	3.14
88	1.50	13	3	2	8.67	2.00	1.33
89	8.50	75	42	27	8.82	4.94	3.18
90	4.00	38	16	9	9.50	4.00	2.25
91	1.50	12	6	3	8.00	4.00	2.00
92	2.00	15	8	7	7.50	4.00	3.50
93	1.50	13	5	3	8.67	3.33	2.00
94	5.50	53	26	13	9.64	4.73	2.36
95	6.00	53	20	17	8.83	3.33	2.83
96	7.00	75	26	18	10.71	3.71	2.57
97	0.00	0	0	0	/	/	/
98	11.00	95	52	33	8.64	4.73	3.00
99	15.00	117	60	47	7.80	4.00	3.13
100	4.50	49	18	13	10.89	4.00	2.89
101	3.50	19	12	8	5.43	3.43	2.29
102	7.50	62	30	21	8.27	4.00	2.80
103	1.00	5	3	1	5.00	3.00	1.00
104	8.50	77	36	23	9.06	4.24	2.71
105	6.50	50	22	16	7.69	3.38	2.46
106	2.50	17	12	7	6.80	4.80	2.80
107	16.50	155	58	38	9.39	3.52	2.30
Mean	9.27	85.17	36.64	23.14	7.94	3.49	2.32
SE	0.86	9.03	4.17	2.13	0.29	0.13	0.09

Table S5

Peak and significant cluster of fMRI activity during SO main effect.

We used the SO main effect whole-brain activation patterns in **Fig. 3b**. ROIs were defined anatomically (see Methods). Cluster sizes are reported $p_{unc.} < 0.001$.

Brain areas	Laterality	Peak T-value	Cluster size	Peak coordinate (MNI)		
				x	y	z
Anterior Supramarginal Gyrus	L	-3.93	303	-52	-29	40
Anterior Supramarginal Gyrus	R	-2.66	18	67	-29	27
Superior Frontal Gyrus	L	-4.53	2766	-12	4	66
Superior Frontal Gyrus	R	-4.86	2122	12	21	59
Middle Frontal Gyrus	L	-5.57	3320	-44	26	33
Middle Frontal Gyrus	R	-5.56	3312	40	25	37
Anterior Paracingulate Gyrus	L	-7.28	4153	-5	38	-10
Anterior Paracingulate Gyrus	R	-7.13	3918	4	45	-8
Angular Gyrus	L	-3.52	519	-62	-54	26
Angular Gyrus	R	-2.66	401	50	-52	49
Frontal Orbital Cortex	L	-6.46	6392	-38	33	-17
Frontal Orbital Cortex	R	-7.26	5637	28	28	-20
Superior Parietal Lobule	L	-5.10	1774	-35	-44	58
Superior Parietal Lobule	R	-5.96	1457	35	-41	59
Insular Cortex	L	-4.89	1004	-32	-26	16
Insular Cortex	R	-6.34	1028	34	-24	9
Posterior Cingulate Gyrus	L	-10.83	4525	-5	-47	19
Posterior Cingulate Gyrus	R	-10.47	5838	5	-43	27
Posterior Supramarginal Gyrus	L	-3.24	102	-61	-51	26
Posterior Supramarginal Gyrus	R	-2.41	7	67	-38	21

Table S6

Peak and significant cluster of fMRI activity during spindle main effect.

We used the Spindle main effect whole-brain activation patterns in **Fig. 3c**. ROIs were defined anatomically (see Methods). Cluster sizes are reported $p_{unc.} < 0.001$.

Brain areas	Laterality	Peak T-value	Cluster size	Peak coordinate (MNI)		
				x	y	z
Anterior Supramarginal Gyrus	L	-6.36	1357	-55	-29	45
Anterior Supramarginal Gyrus	R	-6.53	1671	55	-26	49
Superior Frontal Gyrus	L	-6.83	2274	-20	33	38
Superior Frontal Gyrus	R	5.52	2320	12	-4	75
Middle Frontal Gyrus	L	-8.03	5350	-42	30	30
Middle Frontal Gyrus	R	-5.77	4194	31	28	41
Anterior Paracingulate Gyrus	L	7.15	1086	-7	11	42
Anterior Paracingulate Gyrus	R	6.68	2222	2	15	41
Angular Gyrus	L	-7.55	1240	-48	-58	50
Angular Gyrus	R	-7.33	4588	53	-53	34
Frontal Orbital Cortex	L	-4.61	1149	-41	21	-11
Frontal Orbital Cortex	R	-3.98	645	40	20	-16
Superior Parietal Lobule	L	-6.91	1687	-36	-56	61
Superior Parietal Lobule	R	-7.79	3091	42	-43	62
Insular Cortex	L	-6.99	1103	-37	-20	15
Insular Cortex	R	-7.26	2169	36	-22	12
Posterior Cingulate Gyrus	L	-7.85	3208	-5	-46	38
Posterior Cingulate Gyrus	R	-7.10	3352	0	-45	36
Posterior Supramarginal Gyrus	L	-6.71	1263	-54	-49	39
Posterior Supramarginal Gyrus	R	-6.61	2376	48	-38	54

Brain areas	Laterality	Peak T-value	Cluster size	Peak coordinate (MNI)		
				x	y	z
Anterior Supramarginal Gyrus	L	-4.95	1327	-56	-33	38
Anterior Supramarginal Gyrus	R	-2.42	10	60	-23	27
Superior Frontal Gyrus	L	-3.50	398	-14	10	64
Superior Frontal Gyrus	R	-4.44	596	6	23	49
Middle Frontal Gyrus	L	-4.74	2455	-50	11	40
Middle Frontal Gyrus	R	-6.01	1694	53	17	37
Anterior Paracingulate Gyrus	L	-3.56	381	-1	21	47
Anterior Paracingulate Gyrus	R	-4.98	725	7	21	47
Angular Gyrus	L	1.71	0	-62	-54	26
Angular Gyrus	R	2.63	149	55	-58	19
Frontal Orbital Cortex	L	-4.72	1492	-47	26	-6
Frontal Orbital Cortex	R	-4.24	872	40	27	-2
Superior Parietal Lobule	L	-4.83	1177	-37	-46	56
Superior Parietal Lobule	R	-4.02	584	28	-50	54
Insular Cortex	L	-4.01	1385	-38	19	-1
Insular Cortex	R	-4.36	1374	32	21	0
Posterior Cingulate Gyrus	L	-2.93	83	-9	-32	40
Posterior Cingulate Gyrus	R	-2.91	130	7	-30	42
Posterior Supramarginal Gyrus	L	-1.96	0	-48	-44	46
Posterior Supramarginal Gyrus	R	-2.02	0	49	-37	51

Table S7

Peak and significant cluster of fMRI activity during SO-spindle interaction.

We used the SO-spindle interaction effect whole-brain activation patterns in **Fig. 3d**. ROIs were defined anatomically (see Methods). Cluster sizes are reported $p_{unc.} < 0.001$.

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Reviewer #1 (Public review):

Wang et al., recorded concurrent EEG-fMRI in 107 participants during nocturnal NREM sleep to investigate brain activity and connectivity related to slow oscillations (SO), sleep spindles, and in particular their co-occurrence. The authors found SO-spindle coupling to be correlated with increased thalamic and hippocampal activity, and with increased functional connectivity from the hippocampus to the thalamus and from the thalamus to the neocortex, especially the medial prefrontal cortex (mPFC). They concluded the brain-wide activation pattern to resemble episodic memory processing, but to be dissociated from task-related processing and suggest that the thalamus plays a crucial role in coordinating the hippocampal-cortical dialogue during sleep.

The paper offers an impressively large and highly valuable dataset that provides the opportunity for gaining important new insights into the network substrate involved in SOs, spindles, and their coupling.

Comments on revisions:

While the authors have sufficiently addressed some of my previous comments, I still have severe concerns regarding several key aspects of the methodology, which were even corroborated by the supplementary results presented in response to the last round of reviews. I have the following specific comments (numbers refer to comments raised in the previous review):

Re 1: The revised introduction now cites a couple of papers but discusses them only very superficially, lumping together several studies with very different key results. This is still not very informative for the reader and does not sufficiently acknowledge previously published work. Here are two examples to illustrate this:

- a. "These studies have generally reported that slow oscillations are associated with widespread cortical and subcortical BOLD changes, whereas spindles elicit activation in the thalamus, as well as in several cortical and paralimbic regions." □ Several studies even showed e.g., a clear activation of the hippocampus and parahippocampal gyrus associated with spindles, not just the thalamus
- b. "Although these findings provide valuable insights into the BOLD correlates of sleep rhythms, they often do not employ sophisticated temporal modeling (Huang et al., 2024) [...]." - previous studies have used e.g., spindle event-related regressors with individual spindle amplitudes as parametric modulators, first and second order derivatives of the HRF function, as well as PPI connectivity analyses, which I would consider rather sophisticated temporal modelling.

Re 4+9: The short overall recordings in some subjects on the one hand and the large number of spindles and SOs detected in N1 sleep stages are still highly concerning, in fact even more so, now that the actual numbers have been provided in the Supplementary Tables. Either the sleep staging or the detection of SO and spindle events must be incorrect. I understand that for specific EEG analysis and fMRI modelling purposes sometimes slightly different thresholds are used as compared to clinical sleep staging, but several parameters here are alarmingly off.

- a. Given that proper NREM sleep (N2+N3) is the relevant stage for the analyses conducted in this paper, some of the N2+N3 durations are very short (eg 7-8 min) while those subjects' results have the same impact on the group level analyses as those with >100 min of N2+N3. Either subjects with very little relevant data (not overall recording time but N2+N3 time) should be excluded or weighting subject data for the group analyses according to the amount

od contributed data should be done.

b. The authors argue that the SO and spindle detection algorithms are valid since widely used and that they were developed for N2+N3 stages, which is why they will also detect events in other stages: "While, because the detection methods for SO and spindle are based on percentiles, this method will always detect a certain number of events when used for other stages (N1 and REM) sleep data, but the differences between these events and those detected in stage N23 remain unclear." I do agree that with very liberal thresholds, also SO and spindle events may be detected in other stages, but it shouldn't be that many. If the percentiles of amplitude thresholds were defined based on properly scored N2+N3 stages only, very few events should be detected (erroneously!) in N1, as the occurrence of K-complexes (isolated SOs) and spindles per definition makes it N2, and during REM sleep only very few spindles and SOS are allowed to occur, without scoring it NREM instead. For the first subject (just as example, but with similar numbers for the rest of the sample), reveals as many as 60 SOs and 31 spindles within 8 min of N1 sleep (Table S2) as well as 13 SOs and 7 spindles within 2 min of REM sleep (Table S4). These numbers are completely unrealistic and question the correctness of the sleep staging as well as the physiological relevance of the EEG graphoelements identified as SO and spindles. It also completely undermines the interpretability of the respective event regressors for the fMRI analyses.

c. Likely, given the large numbers of coupled SO-spindle events and the apparently very low amplitude criteria for event identification, also the number of SO-spindle couplings is likely severely overestimated.

Re 10: The rationale for using a lateralized frontal electrode (F3) for both SO (should have been at least bilateral or central) and spindle detection (should have been a centro-parietal electrode) is not convincing. Other EEG-fMRI spindle or SO papers have used a number of frontal (SO) or centro-parietal (spindles) electrodes averaged or even approaches including all EEG electrodes. Searching events with low thresholds at suboptimal recording sites does not do this highly valuable dataset justice.

Re 7: It is not clear to me why/how larger voxels would reduce susceptibility-related distortions and partial volume effects. Usually, the opposite is true. This should be elaborated.

<https://doi.org/10.7554/eLife.103956.2.sa3>

Reviewer #2 (Public review):

In this study, Wang and colleagues aimed to explore brain-wide activation patterns associated with NREM sleep oscillations, including slow oscillations (SOs), spindles, and SO-spindle coupling events. Their findings reveal that SO-spindle events corresponded with increased activation in both the thalamus and hippocampus. Additionally, they observed that SO-spindle coupling was linked to heightened functional connectivity from the hippocampus to the thalamus, and from the thalamus to the medial prefrontal cortex—three key regions involved in memory consolidation and episodic memory processes.

This study's findings are timely and highly relevant to the field. The authors' extensive data collection, involving 107 participants sleeping in an fMRI while undergoing simultaneous EEG recording, deserves special recognition. If shared, this unique dataset could lead to further valuable insights.

Comments on revisions:

The authors' efforts in revising the manuscript and addressing the reviewers' comments are certainly commendable. However, I remain concerned about potential issues in detecting sleep-related oscillations (SOs, spindles, and consequently coupled SO-spindle events), which

may arise due to suboptimal parameter selection or inaccurate sleep staging, potentially impacting all subsequent analyses.

A review of Supplementary Tables 1-4 reveals an unusually high number of detected SOs and spindles during sleep stage N1 and REM sleep. While the authors correctly note that a percentile-based detection approach will always identify a certain number of events across sleep stages, the particularly high counts in N1 and REM are concerning. To mitigate the limitations of this method, the authors could have performed event detection independently of sleep stages (i.e., across the entire dataset for each participant) and subsequently assigned the detected events to the corresponding sleep stages. If the event counts in N1 and REM remained disproportionately high, this would indicate a fundamental issue with the detection procedure.

<https://doi.org/10.7554/eLife.103956.2.sa2>

Reviewer #3 (Public review):

Summary:

Wang et al., examined the brain activity patterns during sleep, especially when locked to those canonical sleep rhythms such as SO, spindle, and their coupling. Analyzing data from a large sample, the authors found significant coupling between spindles and SOs, particularly during the up-state of the SO. Moreover, the authors examined the patterns of whole-brain activity locked to these sleep rhythms. The authors next investigated the functional connectivity analyses, and found enhanced connectivity between the hippocampus and the thalamus and the medial PFC. These results reinforced the theoretical model of sleep-dependent memory consolidation, such that SO-spindle coupling is conducive for systems-level memory reactivation and consolidation.

Strengths:

There are obvious strengths in this work, including the large sample size, state-of-the-art neuroimaging and neural oscillation analyses, and the richness of results. The results now inform hemodynamic neural activity that coincided with SO-spindle couplings.

Weaknesses:

My earlier comments were about the inability to make inferences on memory given the lack of memory tasks, and the weakness in using the open-ended cognitive state decoding.

The current revision has addressed these major concerns. The authors expanded discussions regarding the theoretical implications of the work in a more nuanced manner.

<https://doi.org/10.7554/eLife.103956.2.sa1>

Author response:

The following is the authors' response to the original reviews

Public Reviews:

Reviewer #1 (Public review):

Wang et al., recorded concurrent EEG-fMRI in 107 participants during nocturnal NREM sleep to investigate brain activity and connectivity related to slow oscillations (SO), sleep spindles, and in particular their co-occurrence. The authors found SO-spindle coupling to

be correlated with increased thalamic and hippocampal activity, and with increased functional connectivity from the hippocampus to the thalamus and from the thalamus to the neocortex, especially the medial prefrontal cortex (mPFC). They concluded the brain-wide activation pattern to resemble episodic memory processing, but to be dissociated from task-related processing and suggest that the thalamus plays a crucial role in coordinating the hippocampal-cortical dialogue during sleep.

The paper offers an impressively large and highly valuable dataset that provides the opportunity for gaining important new insights into the network substrate involved in SOs, spindles, and their coupling. However, the paper does unfortunately not exploit the full potential of this dataset with the analyses currently provided, and the interpretation of the results is often not backed up by the results presented. I have the following specific comments.

Thank you for your thoughtful and constructive feedback. We greatly appreciate your recognition of the strengths of our dataset and findings. Below, we address your specific comments and provide responses to each point you raised to ensure our methods and results are as transparent and comprehensible as possible. We hope these revisions address your comments and further strengthen our manuscript. Thank you again for the constructive feedback.

(1) The introduction is lacking sufficient review of the already existing literature on EEG-fMRI during sleep and the BOLD-correlates of slow oscillations and spindles in particular (Laufs et al., 2007; Schabus et al., 2007; Horovitz et al., 2008; Laufs, 2008; Czigic et al., 2009; Picchioni et al., 2010; Spoormaker et al., 2010; Caporro et al., 2011; Bergmann et al., 2012; Hale et al., 2016; Fogel et al., 2017; Moehlmann et al., 2018; Ilhan-Bayrakci et al., 2022). The few studies mentioned are not discussed in terms of the methods used or insights gained.

We acknowledge the need for a more comprehensive review of prior EEG-fMRI studies investigating BOLD correlates of slow oscillations and spindles. However, these articles are not all related to sleep SO or spindle. Articles (Hale et al., 2016; Horovitz et al., 2008; Laufs, 2008; Laufs, Walker, & Lund, 2007; Spoormaker et al., 2010) mainly focus on methodology for EEG-fMRI, sleep stages, or brain networks, which are not the focus of our study. Thank you again for your attention to the comprehensiveness of our literature review, and we will expand the introduction to include a more detailed discussion of the existing literature, ensuring that the contributions of previous EEG-fMRI sleep studies are adequately acknowledged.

Introduction, Page 4 Lines 62-76

“Investigating these sleep-related neural processes in humans is challenging because it requires tracking transient sleep rhythms while simultaneously assessing their widespread brain activation. Recent advances in simultaneous EEG-fMRI techniques provide a unique opportunity to explore these processes. EEG allows for precise event-based detection of neural signal, while fMRI provides insight into the broader spatial patterns of brain activation and functional connectivity (Horovitz et al., 2008; Huang et al., 2024; Laufs, 2008; Laufs, Walker, & Lund, 2007; Schabus et al., 2007; Spoormaker et al., 2010). Previous EEG-fMRI studies on sleep have focused on classifying sleep stages or examining the neural correlates of specific waves (Bergmann et al., 2012; Caporro et al., 2012; Czigic et al., 2009; Fogel et al., 2017; Hale et al., 2016; Ilhan-Bayrakci et al., 2022; Moehlmann et al., 2019; Picchioni et al., 2011). These studies have generally reported that slow oscillations are associated with widespread cortical and subcortical BOLD changes, whereas spindles elicit activation in the thalamus, as well as in several cortical and paralimbic regions. Although these findings provide valuable insights into the BOLD correlates of sleep rhythms, they often do not employ

sophisticated temporal modeling (Huang et al., 2024), to capture the dynamic interactions between different oscillatory events, e.g., the coupling between SOs and spindles.”

(2) The paper falls short in discussing the specific insights gained into the neurobiological substrate of the investigated slow oscillations, spindles, and their interactions. The validity of the inverse inference approach (“Open ended cognitive state decoding”), assuming certain cognitive functions to be related to these oscillations because of the brain regions/networks activated in temporal association with these events, is debatable at best. It is also unclear why eventually only episodic memory processing-like brain-wide activation is discussed further, despite the activity of 16 of 50 feature terms from the NeuroSynth v3 dataset were significant (episodic memory, declarative memory, working memory, task representation, language, learning, faces, visuospatial processing, category recognition, cognitive control, reading, cued attention, inhibition, and action).

Thank you for pointing this out, particularly regarding the use of inverse inference approaches such as “open-ended cognitive state decoding.” Given the concerns about the indirectness of this approach, we decided to remove its related content and results from Figure 3 in the main text and include it in Supplementary Figure 7. We will refocus the main text on direct neurobiological insights gained from our EEG-fMRI analyses, particularly emphasizing the hippocampal-thalamocortical network dynamics underlying SO-spindle coupling, and we will acknowledge the exploratory nature of these findings and highlight their limitations.

Discussion, Page 17-18 Lines 323-332

“To explore functional relevance, we employed an open-ended cognitive state decoding approach using meta-analytic data (NeuroSynth: Yarkoni et al. (2011)). Although this method usefully generates hypotheses about potential cognitive processes, particularly in the absence of a pre- and post-sleep memory task, it is inherently indirect. Many cognitive terms showed significant associations (16 of 50), such as “episodic memory,” “declarative memory,” and “working memory.” We focused on episodic/declarative memory given the known link with hippocampal reactivation (Diekelmann & Born, 2010; Staresina et al., 2015; Staresina et al., 2023). Nonetheless, these inferences regarding memory reactivation should be interpreted cautiously without direct behavioral measures. Future research incorporating explicit tasks before and after sleep would more rigorously validate these potential functional claims.”

(3) Hippocampal activation during SO-spindles is stated as a main hypothesis of the paper - for good reasons - however, other regions (e.g., several cortical as well as thalamic) would be equally expected given the known origin of both oscillations and the existing sleep-EEG-fMRI literature. However, this focus on the hippocampus contrasts with the focus on investigating the key role of the thalamus instead in the Results section.

We appreciate your insight regarding the relative emphasis on hippocampal and thalamic activation in our study. We recognize that the manuscript may currently present an inconsistency between our initial hypothesis and the main focus of the results. To address this concern, we will ensure that our Introduction and Discussion section explicitly discusses both regions, highlighting the complementary roles of the hippocampus (memory processing and reactivation) and the thalamus (spindle generation and cortico-hippocampal coordination) in SO-spindle dynamics.

Introduction, Page 5 Lines 87-103

“To address this gap, our study investigates brain-wide activation and functional connectivity patterns associated with SO-spindle coupling, and employs a cognitive state decoding approach (Margulies et al., 2016; Yarkoni et al., 2011)—albeit indirectly—to infer potential

cognitive functions. In the current study, we used simultaneous EEG-fMRI recordings during nocturnal naps (detailed sleep staging results are provided in the Methods and Table S1) in 107 participants. Although directly detecting hippocampal ripples using scalp EEG or fMRI is challenging, we expected that hippocampal activation in fMRI would coincide with SO-spindle coupling detected by EEG, given that SOs, spindles, and ripples frequently co-occur during NREM sleep. We also anticipated a critical role of the thalamus, particularly thalamic spindles, in coordinating hippocampal-cortical communication.

We found significant coupling between SOs and spindles during NREM sleep (N2/3), with spindle peaks occurring slightly before the SO peak. This coupling was associated with increased activation in both the thalamus and hippocampus, with functional connectivity patterns suggesting thalamic coordination of hippocampal-cortical communication. These findings highlight the key role of the thalamus in coordinating hippocampal-cortical interactions during human sleep and provide new insights into the neural mechanisms underlying sleep-dependent brain communication. A deeper understanding of these mechanisms may contribute to future neuromodulation approaches aimed at enhancing sleep-dependent cognitive function and treating sleep-related disorders.”

Discussion, Page 16-17 Lines 292-307

“When modeling the timing of these sleep rhythms in the fMRI, we observed hippocampal activation selectively during SO-spindle events. This suggests the possibility of triple coupling (SOs–spindles–ripples), even though our scalp EEG was not sufficiently sensitive to detect hippocampal ripples—key markers of memory replay (Buzsáki, 2015). Recent iEEG evidence indicates that ripples often co-occur with both spindles (Ngo, Fell, & Staresina, 2020) and SOs (Staresina et al., 2015; Staresina et al., 2023). Therefore, the hippocampal involvement during SO-spindle events in our study may reflect memory replay from the hippocampus, propagated via thalamic spindles to distributed cortical regions.

The thalamus, known to generate spindles (Halassa et al., 2011), plays a key role in producing and coordinating sleep rhythms (Coulon, Budde, & Pape, 2012; Crunelli et al., 2018), while the hippocampus is found essential for memory consolidation (Buzsáki, 2015; Diba & Buzsáki, 2007; Singh, Norman, & Schapiro, 2022). The increased hippocampal and thalamic activity, along with strengthened connectivity between these regions and the mPFC during SO-spindle events, underscores a hippocampal-thalamic-neocortical information flow. This aligns with recent findings suggesting the thalamus orchestrates neocortical oscillations during sleep (Schreiner et al., 2022). The thalamus and hippocampus thus appear central to memory consolidation during sleep, guiding information transfer to the neocortex, e.g., mPFC.”

(4) The study included an impressive number of 107 subjects. It is surprising though that only 31 subjects had to be excluded under these difficult recording conditions, especially since no adaptation night was performed. Since only subjects were excluded who slept less than 10 min (or had excessive head movements) there are likely several datasets included with comparably short durations and only a small number of SOs and spindles and even less combined SO-spindle events. A comprehensive table should be provided (supplement) including for each subject (included and excluded) the duration of included NREM sleep, number of SOs, spindles, and SO+spindle events. Also, some descriptive statistics (mean/SD/range) would be helpful.

We appreciate your recognition of our sample size and the challenges associated with simultaneous EEG-fMRI sleep recordings. We acknowledge the importance of transparently reporting individual subject data, particularly regarding sleep duration and the number of detected SOs, spindles, and SO-spindle events. To address this, we will provide comprehensive tables in the supplementary materials, contains descriptive information about sleep-related characteristics (Table S1), as well as detailed information about sleep

waves at each sleep stage for all 107 subjects (Table S2-S4), listing for each subject: (1) Different sleep stage duration; (2) Number of detected SOs; (3) Number of detected spindles; (4) Number of detected SO-spindle coupling events; (5) Density of detected SOs; (6) Density of detected spindles; (7) Density of detected SO-spindle coupling events.

However, most of the excluded participants were unable to fall asleep or had too short a sleep duration, so they basically had no NREM sleep period, so it was impossible to count the NREM sleep duration, SO, spindle, and coupling numbers.

Supplementary Materials, Page 42-54, Table S1-S4

(5) Was the 20-channel head coil dedicated for EEG-fMRI measurements? How were the electrode cables guided through/out of the head coil? Usually, the 64-channel head coil is used for EEG-fMRI measurements in a Siemens PRISMA 3T scanner, which has a cable duct at the back that allows to guide the cables straight out of the head coil (to minimize MR-related artifacts). The choice for the 20-channel head coil should be motivated. Photos of the recording setup would also be helpful.

Thank you for your comment regarding our choice of the 20-channel head coil for EEG-fMRI measurements. We acknowledge that the 64-channel head coil is commonly used in Siemens PRISMA 3T scanners; however, the 20-channel coil was selected due to specific practical and technical considerations in our study. In particular, the 20-channel head coil was compatible with our EEG system and ensured sufficient signal-to-noise ratio (SNR) for both EEG and fMRI acquisition. The EEG electrode cables were guided through the lateral and posterior openings of the head coil, secured with foam padding to reduce motion and minimize MR-related artifacts. Moreover, given the extended nature of nocturnal sleep recordings, the 20-channel coil allowed us to maintain participant comfort while still achieving high-quality simultaneous EEG-fMRI data.

We have made this clearer in the revised manuscript.

Methods, Page 20 Lines 385-392

“All MRI data were acquired using a 20-channel head coil on a research-dedicated 3-Tesla Siemens Magnetom Prisma MRI scanner. Earplugs and cushions were provided for noise protection and head motion restriction. We chose the 20-channel head coil because it was compatible with our EEG system and ensured sufficient signal-to-noise ratio (SNR) for both EEG and fMRI acquisition. The EEG electrode cables were guided through the lateral and posterior openings of the head coil, secured with foam padding to reduce motion and minimize MR-related artifacts. Moreover, given the extended nature of nocturnal sleep recordings, the 20-channel coil helped maintain participant comfort while still achieving high-quality simultaneous EEG-fMRI data.”

(6) Was the EEG sampling synchronized to the MR scanner (gradient system) clock (the 10 MHz signal; not referring to the volume TTL triggers here)? This is a requirement for stable gradient artifact shape over time and thus accurate gradient noise removal.

Thank you for raising this important point. We confirm that the EEG sampling was synchronized to the MR scanner's 10 MHz gradient system clock, ensuring a stable gradient artifact shape over time and enabling accurate artifact removal. This synchronization was achieved using the standard clock synchronization interface of the EEG amplifier, minimizing timing jitter and drift. As a result, the gradient artifact waveform remained stable across volumes, allowing for more effective artifact correction during preprocessing. We appreciate your attention to this critical aspect of EEG-fMRI data acquisition.

We have made this clearer in the revised manuscript.

Methods, Page 19-20 Lines 371-383

“EEG was recorded simultaneously with fMRI data using an MR-compatible EEG amplifier system (BrainAmps MR-Plus, Brain Products, Germany), along with a specialized electrode cap. The recording was done using 64 channels in the international 10/20 system, with the reference channel positioned at FCz. In order to adhere to polysomnography (PSG) recording standards, six electrodes were removed from the EEG cap: one for electrocardiogram (ECG) recording, two for electrooculogram (EOG) recording, and three for electromyogram (EMG) recording. EEG data was recorded at a sample rate of 5000 Hz, the resistance of the reference and ground channels was kept below 10 k Ω , and the resistance of the other channels was kept below 20 k Ω . To synchronize the EEG and fMRI recordings, the BrainVision recording software (BrainProducts, Germany) was utilized to capture triggers from the MRI scanner. The EEG sampling was synchronized to the MR scanner’s 10 MHz gradient system clock, ensuring a stable gradient artifact shape over time and enabling accurate artifact removal. This was achieved via the standard clock synchronization interface of the EEG amplifier, minimizing timing jitter and drift.”

(7) The TR is quite long and the voxel size is quite large in comparison to state-of-the-art EPI sequences. What was the rationale behind choosing a sequence with relatively low temporal and spatial resolution?

We acknowledge that our chosen TR and voxel size are relatively long and large compared to state-of-the-art EPI sequences. This decision was made to optimize the signal-to-noise ratio (SNR) and reduce susceptibility-related distortions, which are particularly critical in EEG-fMRI sleep studies where head motion and physiological noise can be substantial. A longer TR allowed us to sample whole-brain activity with sufficient coverage, while a larger voxel size helped enhance BOLD sensitivity and minimize partial volume effects in deep brain structures such as the thalamus and hippocampus, which are key regions of interest in our study. We appreciate your concern and hope this clarification provides sufficient rationale for our sequence parameters.

We have made this clearer in the revised manuscript.

Methods, Page 20-21 Lines 398-408

“Then, the “sleep” session began after the participants were instructed to try and fall asleep. For the functional scans, whole-brain images were acquired using k-space and steady-state T2*-weighted gradient echo-planar imaging (EPI) sequence that is sensitive to the BOLD contrast. This measures local magnetic changes caused by changes in blood oxygenation that accompany neural activity (sequence specification: 33 slices in interleaved ascending order, TR = 2000 ms, TE = 30 ms, voxel size = 3.5 × 3.5 × 4.2 mm³, FA = 90°, matrix = 64 × 64, gap = 0.7 mm). A relatively long TR and larger voxel size were chosen to optimize SNR and reduce susceptibility-related distortions, which are critical in EEG-fMRI sleep studies where head motion and physiological noise can be substantial. The longer TR allowed whole-brain coverage with sufficient temporal resolution, while the larger voxel size helped enhance BOLD sensitivity and minimize partial volume effects in deep brain structures (e.g., the thalamus and hippocampus), which are key regions of interest in this study.”

(8) The anatomically defined ROIs are quite large. It should be elaborated on how this might reduce sensitivity to sleep rhythm-specific activity within sub-regions, especially for the thalamus, which has distinct nuclei involved in sleep functions.

We appreciate your insight regarding the use of anatomically defined ROIs and their potential limitations in detecting sleep rhythm-specific activity within sub-regions, particularly in the thalamus. Given the distinct functional roles of thalamic nuclei in sleep

processes, we acknowledge that using a single, large thalamic ROI may reduce sensitivity to localized activity patterns. To address this, we will discuss this limitation in the revised manuscript, acknowledging that our approach prioritizes whole-structure effects but may not fully capture nucleus-specific contributions.

Discussion, Page 18 Lines 333-341

“Despite providing new insights, our study has several limitations. First, our scalp EEG did not directly capture hippocampal ripples, preventing us from conclusively demonstrating triple coupling. Second, the combination of EEG-fMRI and the lack of a memory task limit our ability to parse fine-grained BOLD responses at the DOWN- vs. UP-states of SOs and link observed activations to behavioral outcomes. Third, the use of large anatomical ROIs may mask subregional contributions of specific thalamic nuclei or hippocampal subfields. Finally, without a memory task, we cannot establish a direct behavioral link between sleep-rhythm-locked activation and memory consolidation. Future studies combining techniques such as ultra-high-field fMRI or iEEG with cognitive tasks may refine our understanding of subregional network dynamics and functional significance during sleep.”

(9) The study reports SO & spindle amplitudes & densities, as well as SO+spindle coupling, to be larger during N2/3 sleep compared to N1 and REM sleep, which is trivial but can be seen as a sanity check of the data. However, the amount of SOs and spindles reported for N1 and REM sleep is concerning, as per definition there should be hardly any (if SOs or spindles occur in N1 it becomes by definition N2, and the interval between spindles has to be considerably large in REM to still be scored as such). Thus, on the one hand, the report of these comparisons takes too much space in the main manuscript as it is trivial, but on the other hand, it raises concerns about the validity of the scoring.

We appreciate your concern regarding the reported presence of SOs and spindles in N1 and REM sleep and the potential implications. Our detection method for detecting SO, spindle, and coupling were originally designed only for N2&N3 sleep data based on the characteristics of the data itself, and this method is widely recognized and used in the sleep research (Hahn et al., 2020; Helfrich et al., 2019; Helfrich et al., 2018; Ngo, Fell, & Staresina, 2020; Schreiner et al., 2022; Schreiner et al., 2021; Staresina et al., 2015; Staresina et al., 2023). While, because the detection methods for SO and spindle are based on percentiles, this method will always detect a certain number of events when used for other stages (N1 and REM) sleep data, but the differences between these events and those detected in stage N2/3 remain unclear. We will acknowledge the reasons for these results in the Methods section and emphasize that they are used only for sanity checks.

Methods, Page 25 Lines 515-524

“We note that the above methods for detecting SOs, spindles, and their couplings were originally developed for N2 and N3 sleep data, based on the specific characteristics of these stages. These methods are widely recognized in sleep research (Hahn et al., 2020; Helfrich et al., 2019; Helfrich et al., 2018; Ngo, Fell, & Staresina, 2020; Schreiner et al., 2022; Schreiner et al., 2021; Staresina et al., 2015; Staresina et al., 2023). However, because this percentile-based detection approach will inherently identify a certain number of events if applied to other stages (e.g., N1 and REM), the nature of these events in those stages remains unclear compared to N2/N3. We nevertheless identified and reported the detailed descriptive statistics of these sleep rhythms in all sleep stages, under the same operational definitions, both for completeness and as a sanity check. Within the same subject, there should be more SOs, spindles, and their couplings in N2/N3 than in N1 or REM (see also Figure S2-S4, Table S1-S4).”

(10) Why was electrode F3 used to quantify the occurrence of SOs and spindles? Why not a midline frontal electrode like Fz (or a number of frontal electrodes for SOs) and Cz (or a number of centroparietal electrodes) for spindles to be closer to their maximum topography?

We appreciate your suggestion regarding electrode selection for SO and spindle quantification. Our choice of F3 was primarily based on previous studies (Massimini et al., 2004; Molle et al., 2011), where bilateral frontal electrodes are commonly used for detecting SOs and spindles. Additionally, we considered the impact of MRI-related noise and, after a comprehensive evaluation, determined that F3 provided an optimal balance between signal quality and artifact minimization. We also acknowledge that alternative electrode choices, such as Fz for SOs and Cz for spindles, could provide additional insights into their topographical distributions.

(11) Functional connectivity (hippocampus -> thalamus -> cortex (mPFC)) is reported to be increased during SO-spindle coupling and interpreted as evidence for coordination of hippocampo-neocortical communication likely by thalamic spindles. However, functional connectivity was only analysed during coupled SO+spindle events, not during isolated SOs or isolated spindles. Without the direct comparison of the connectivity patterns between these three events, it remains unclear whether this is specific for coupled SO+spindle events or rather associated with one or both of the other isolated events. The PPIs need to be conducted for those isolated events as well and compared statistically to the coupled events.

We appreciate your critical perspective on our functional connectivity analysis and the interpretation of hippocampus-thalamus-cortex (mPFC) interactions during SO-spindle coupling. We acknowledge that, in the current analysis, functional connectivity was only examined during coupled SO-spindle events, without direct comparison to isolated SOs or isolated spindles. To address this concern, we have conducted PPI analyses for all three ROIs (Hippocampus, Thalamus, mPFC) and all three event types (SO-spindle couplings, isolated SOs, and isolated spindles). Our results indicate that neither isolated SOs nor isolated spindles yielded significant connectivity changes in all three ROIs, as all failed to survive multiple comparison corrections. This suggests that the observed connectivity increase is specific to SO-spindle coupling, rather than being independently driven by either SOs or spindles alone.

Results, Page 14 Lines 248-255

“Crucially, the interaction between FC and SO-spindle coupling revealed that only the functional connectivity of hippocampus -> thalamus (ROI analysis, $t(106) = 1.86$, $p = 0.0328$) and thalamus -> mPFC (ROI analysis, $t(106) = 1.98$, $p = 0.0251$) significantly increased during SO-spindle coupling, with no significant changes in all other pathways (Fig. 4e). We also conducted PPI analyses for the other two events (SOs and spindles), and neither yielded significant connectivity changes in the three ROIs, as all failed to survive whole-brain FWE correction at the cluster level ($p < 0.05$). Together, these findings suggest that the thalamus, likely via spindles, coordinates hippocampal-cortical communication selectively during SO-spindle coupling, but not isolated SOs or spindle events alone.”

(12) The limited temporal resolution of fMRI does indeed not allow for easily distinguishing between fMRI activation patterns related to SO-up- vs. SO-down-states. For this, one could try to extract the amplitudes of SO-up- and SO-down-states separately for each SO event and model them as two separate parametric modulators (with the risk of collinearity as they are likely correlated).

We appreciate your insightful comment regarding the challenge of distinguishing fMRI activation patterns related to SO-up vs. SO-down states due to the limited temporal resolution of fMRI. While our current analysis does not differentiate between these two phases, we acknowledge that separately modeling SO-up and SO-down states using parametric modulators could provide a more refined understanding of their distinct neural correlates. However, as you notes, this approach carries the risk of collinearity, and there is indeed a high correlation between the two amplitudes across all subjects in our results ($r=0.98$). Future studies could explore more on leveraging high-temporal-resolution techniques. While implementing this in the current study is beyond our scope, we will acknowledge this limitation in the Discussion section.

Discussion, Page 17 Lines 308-322

“An intriguing aspect of our findings is the reduced DMN activity during SOs when modeled at the SO trough (DOWN-state). This reduced DMN activity may reflect large-scale neural inhibition characteristic of the SO trough. The DMN is typically active during internally oriented cognition (e.g., self-referential processing or mind-wandering) and is suppressed during external stimuli processing (Yeshurun, Nguyen, & Hasson, 2021). It is unlikely, however, that this suppression of DMN during SO events is related to a shift from internal cognition to external responses given it is during deep sleep time. Instead, it could be driven by the inherent rhythmic pattern of SOs, which makes it difficult to separate UP- from DOWN-states (the two temporal regressors were highly correlated, and similar brain activation during SOs events was obtained if modelled at the SO peak instead, Fig. S5). Since the amplitude at the SO trough is consistently larger than that at the SO peak, the neural activation we detected may primarily capture the large-scale inhibition from DOWN-state. Interestingly, no such DMN reduction was found during SO-spindle coupling, implying that coupling may involve distinct neural dynamics that partially re-engage DMN-related processes, possibly reflecting memory-related reactivation. Future research using high-temporal-resolution techniques like iEEG could clarify these possibilities.”

Discussion, Page 18 Lines 333-341

“Despite providing new insights, our study has several limitations. First, our scalp EEG did not directly capture hippocampal ripples, preventing us from conclusively demonstrating triple coupling. Second, the combination of EEG-fMRI and the lack of a memory task limit our ability to parse fine-grained BOLD responses at the DOWN- vs. UP-states of SOs and link observed activations to behavioral outcomes. Third, the use of large anatomical ROIs may mask subregional contributions of specific thalamic nuclei or hippocampal subfields. Finally, without a memory task, we cannot establish a direct behavioral link between sleep-rhythm-locked activation and memory consolidation. Future studies combining techniques such as ultra-high-field fMRI or iEEG with cognitive tasks may refine our understanding of subregional network dynamics and functional significance during sleep.”

(13) L327: *"It is likely that our findings of diminished DMN activity reflect brain activity during the SO DOWN-state, as this state consistently shows higher amplitude compared to the UP-state within subjects, which is why we modelled the SO trough as its onset in the fMRI analysis." This conclusion is not justified as the fact that SO down-states are larger in amplitude does not mean their impact on the BOLD response is larger.*

We appreciate your concern regarding our interpretation of diminished DMN activity reflecting the SO down-state. We acknowledge that the current expression is somewhat misleading, and our interpretation of it is: it could be driven by the inherent rhythmic pattern of SOs, which makes it difficult to separate UP- from DOWN-states (the two temporal regressors were highly correlated, and similar brain activation during SOs events was obtained if modelled at the SO peak instead). Since the amplitude at the SO trough is

consistently larger than that at the SO peak, the neural activation we detected may primarily capture the large-scale inhibition from DOWN-state. And we will make this clear in the Discussion section.

Discussion, Page 17 Lines 308-322

“An intriguing aspect of our findings is the reduced DMN activity during SOs when modeled at the SO trough (DOWN-state). This reduced DMN activity may reflect large-scale neural inhibition characteristic of the SO trough. The DMN is typically active during internally oriented cognition (e.g., self-referential processing or mind-wandering) and is suppressed during external stimuli processing (Yeshurun, Nguyen, & Hasson, 2021). It is unlikely, however, that this suppression of DMN during SO events is related to a shift from internal cognition to external responses given it is during deep sleep time. Instead, it could be driven by the inherent rhythmic pattern of SOs, which makes it difficult to separate UP- from DOWN-states (the two temporal regressors were highly correlated, and similar brain activation during SOs events was obtained if modelled at the SO peak instead, Fig. S5). Since the amplitude at the SO trough is consistently larger than that at the SO peak, the neural activation we detected may primarily capture the large-scale inhibition from DOWN-state. Interestingly, no such DMN reduction was found during SO-spindle coupling, implying that coupling may involve distinct neural dynamics that partially re-engage DMN-related processes, possibly reflecting memory-related reactivation. Future research using high-temporal-resolution techniques like iEEG could clarify these possibilities.”

(14) Line 77: *"In the current study, while directly capturing hippocampal ripples with scalp EEG or fMRI is difficult, we expect to observe hippocampal activation in fMRI whenever SOs-spindles coupling is detected by EEG, if SOs- spindles-ripples triple coupling occurs during human NREM sleep". Not all SO-spindle events are associated with ripples (Staresina et al., 2015), but hippocampal activation may also be expected based on the occurrence of spindles alone (Bergmann et al., 2012).*

We appreciate your clarification regarding the relationship between SO-spindle coupling and hippocampal ripples. We acknowledge that not all SO-spindle events are necessarily accompanied by ripples (Staresina et al., 2015). However, based on previous research, we found that hippocampal ripples are significantly more likely to occur during SO-spindle coupling events. This suggests that while ripple occurrence is not guaranteed, SO-spindle coupling creates a favorable network state for ripple generation and potential hippocampal activation. To ensure accuracy, we will revise the manuscript to delete this misleading sentence in the Introduction section and acknowledge in the Discussion that our results cannot conclusively directly observe the triple coupling of SO, spindle, and hippocampal ripples.

Discussion, Page 18 Lines 333-341

“Despite providing new insights, our study has several limitations. First, our scalp EEG did not directly capture hippocampal ripples, preventing us from conclusively demonstrating triple coupling. Second, the combination of EEG-fMRI and the lack of a memory task limit our ability to parse fine-grained BOLD responses at the DOWN- vs. UP-states of SOs and link observed activations to behavioral outcomes. Third, the use of large anatomical ROIs may mask subregional contributions of specific thalamic nuclei or hippocampal subfields. Finally, without a memory task, we cannot establish a direct behavioral link between sleep-rhythm-locked activation and memory consolidation. Future studies combining techniques such as ultra-high-field fMRI or iEEG with cognitive tasks may refine our understanding of subregional network dynamics and functional significance during sleep.”

Reviewer #2 (Public review):

In this study, Wang and colleagues aimed to explore brain-wide activation patterns associated with NREM sleep oscillations, including slow oscillations (SOs), spindles, and SO-spindle coupling events. Their findings reveal that SO-spindle events corresponded with increased activation in both the thalamus and hippocampus. Additionally, they observed that SO-spindle coupling was linked to heightened functional connectivity from the hippocampus to the thalamus, and from the thalamus to the medial prefrontal cortex—three key regions involved in memory consolidation and episodic memory processes.

This study's findings are timely and highly relevant to the field. The authors' extensive data collection, involving 107 participants sleeping in an fMRI while undergoing simultaneous EEG recording, deserves special recognition. If shared, this unique dataset could lead to further valuable insights. While the conclusions of the data seem overall well supported by the data, some aspects with regard to the detection of sleep oscillations need clarification.

The authors report that coupled SO-spindle events were most frequent during NREM sleep (2.46 [plus minus] 0.06 events/min), but they also observed a surprisingly high occurrence of these events during N1 and REM sleep (2.23 [plus minus] 0.09 and 2.32 [plus minus] 0.09 events/min, respectively), where SO-spindle coupling would not typically be expected. Combined with the relatively modest SO amplitudes reported (~25 μ V, whereas >75 μ V would be expected when using mastoids as reference electrodes), this raises the possibility that the parameters used for event detection may not have been conservative enough - or that sleep staging was inaccurately performed. This issue could present a significant challenge, as the fMRI findings are largely dependent on the reliability of these detected events.

Thank you very much for your thorough and encouraging review. We appreciate your recognition of the significance and relevance of our study and dataset, particularly in highlighting how simultaneous EEG-fMRI recordings can provide complementary insights into the temporal dynamics of neural oscillations and their associated spatial activation patterns during sleep. In the sections that follow, we address each of your comments in detail. We have revised the text and conducted additional analyses wherever possible to strengthen our argument, clarify our methodological choices. We believe these revisions improve the clarity and rigor of our work, and we thank you for helping us refine it.

We appreciate your insightful comments regarding the detection of sleep oscillations. Our methods for detecting SOs, spindles, and their couplings were originally developed for N2 and N3 sleep data, based on the specific characteristics of these stages. These methods are widely recognized in sleep research (Hahn et al., 2020; Helfrich et al., 2019; Helfrich et al., 2018; Ngo, Fell, & Staresina, 2020; Schreiner et al., 2022; Schreiner et al., 2021; Staresina et al., 2015; Staresina et al., 2023). However, because this percentile-based detection approach will inherently identify a certain number of events if applied to other stages (e.g., N1 and REM), the nature of these events in those stages remains unclear compared to N2/N3. We nevertheless identified and reported the detailed descriptive statistics of these sleep rhythms in all sleep stages, under the same operational definitions, both for completeness and as a sanity check. Within the same subject, there should be more SOs, spindles, and their couplings in N2/N3 than in N1 or REM. We will acknowledge the reasons for these results in the Methods section and emphasize that they are used only for sanity checks.

Regarding the reported SO amplitudes (~25 μ V), during preprocessing, we applied the Signal Space Projection (SSP) method to more effectively remove MRI gradient artifacts and cardiac pulse noise. While this approach enhances data quality, it also reduces overall signal power,

leading to systematically lower reported amplitudes. Despite this, our SO detection in NREM sleep (especially N2/N3) remain physiologically meaningful and are consistent with previous fMRI studies using similar artifact removal techniques. We appreciate your careful evaluation and valuable suggestions.

In addition, we will provide comprehensive tables in the supplementary materials, contains descriptive information about sleep-related characteristics (Table S1), as well as detailed information about sleep waves at each sleep stage for all 107 subjects (Table S2-S4), listing for each subject: (1) Different sleep stage duration; (2) Number of detected SOs; (3) Number of detected spindles; (4) Number of detected SO-spindle coupling events; (2) Density of detected SOs; (3) Density of detected spindles; (4) Density of detected SO-spindle coupling events.

Methods, Page 25 Lines 515-524

“We note that the above methods for detecting SOs, spindles, and their couplings were originally developed for N2 and N3 sleep data, based on the specific characteristics of these stages. These methods are widely recognized in sleep research (Hahn et al., 2020; Helfrich et al., 2019; Helfrich et al., 2018; Ngo, Fell, & Staresina, 2020; Schreiner et al., 2022; Schreiner et al., 2021; Staresina et al., 2015; Staresina et al., 2023). However, because this percentile-based detection approach will inherently identify a certain number of events if applied to other stages (e.g., N1 and REM), the nature of these events in those stages remains unclear compared to N2/N3. We nevertheless identified and reported the detailed descriptive statistics of these sleep rhythms in all sleep stages, under the same operational definitions, both for completeness and as a sanity check. Within the same subject, there should be more SOs, spindles, and their couplings in N2/N3 than in N1 or REM (see also Figure S2-S4, Table S1-S4).”

Supplementary Materials, Page 42-54, Table S1-S4

Reviewer #3 (Public review):

Summary:

Wang et al., examined the brain activity patterns during sleep, especially when locked to those canonical sleep rhythms such as SO, spindle, and their coupling. Analyzing data from a large sample, the authors found significant coupling between spindles and SOs, particularly during the upstate of the SO. Moreover, the authors examined the patterns of whole-brain activity locked to these sleep rhythms. To understand the functional significance of these brain activities, the authors further conducted open-ended cognitive state decoding and found a variety of cognitive processing may be involved during SO-spindle coupling and during other sleep events. The authors next investigated the functional connectivity analyses and found enhanced connectivity between the hippocampus, the thalamus, and the medial PFC. These results reinforced the theoretical model of sleep-dependent memory consolidation, such that SO-spindle coupling is conducive to systems-level memory reactivation and consolidation.

Strengths:

There are obvious strengths in this work, including the large sample size, state-of-the-art neuroimaging and neural oscillation analyses, and the richness of results.

Weaknesses:

Despite these strengths and the insights gained, there are weaknesses in the design, the analyses, and inferences.

Thank you for your detailed and thoughtful review of our manuscript. We are delighted that you recognize our advanced analysis methods and rich results of neuroimaging and neural oscillations as well as the large sample size data. In the following sections, we provide detailed responses to each of your comments. And we have revised the text and conducted additional analyses to strengthen our arguments and clarify our methodological choices. We believe these revisions enhance the clarity and rigor of our work, and we sincerely appreciate your thoughtful feedback in helping us refine the manuscript.

(1) A repeating statement in the manuscript is that brain activity could indicate memory reactivation and thus consolidation. This is indeed a highly relevant question that could be informed by the current data/results. However, an inherent weakness of the design is that there is no memory task before and after sleep. Thus, it is difficult (if not impossible) to make a strong argument linking SO/spindle/coupling-locked brain activity with memory reactivation or consolidation.

We appreciate your suggestion regarding the lack of a pre- and post-sleep memory task in our study design. We acknowledge that, in the absence of behavioral measures, it is hard to directly link SO-spindle coupling to memory consolidation in an outcome-driven manner. Our interpretation is instead based on the well-established role of these oscillations in memory processes, as demonstrated in previous studies. We sincerely appreciate this feedback and will adjust our Discussion accordingly to reflect a more precise interpretation of our findings.

Discussion, Page 18 Lines 333-341

“Despite providing new insights, our study has several limitations. First, our scalp EEG did not directly capture hippocampal ripples, preventing us from conclusively demonstrating triple coupling. Second, the combination of EEG-fMRI and the lack of a memory task limit our ability to parse fine-grained BOLD responses at the DOWN- vs. UP-states of SOs and link observed activations to behavioral outcomes. Third, the use of large anatomical ROIs may mask subregional contributions of specific thalamic nuclei or hippocampal subfields. Finally, without a memory task, we cannot establish a direct behavioral link between sleep-rhythm-locked activation and memory consolidation. Future studies combining techniques such as ultra-high-field fMRI or iEEG with cognitive tasks may refine our understanding of subregional network dynamics and functional significance during sleep.”

(2) Relatedly, to understand the functional implications of the sleep rhythm-locked brain activity, the authors employed the "open-ended cognitive state decoding" method. While this method is interesting, it is rather indirect given that there were no behavioral indices in the manuscript. Thus, discussions based on these analyses are speculative at best. Please either tone down the language or find additional evidence to support these claims.

Moreover, the results from this method are difficult to understand. Figure 3e showed that for all three types of sleep events (SO, spindle, SO-spindle), the same mental states (e.g., working memory, episodic memory, declarative memory) showed opposite directions of activation (left and right panels showed negative and positive activation, respectively). How to interpret these conflicting results? This ambiguity is also reflected by the term used: declarative memory and episodic memories are both indexed in the results. Yet these two processes can be largely overlapped. So which specific memory processes do these brain activity patterns reflect? The Discussion shall discuss these results and the limitations of this method.

We appreciate your critical assessment of the open-ended cognitive state decoding method and its interpretational challenges. Given the concerns about the indirectness of this

approach, we decided to remove its related content and results from Figure 3 in the main text and include it in Supplementary Figure 7.

Due to the complexity of memory-related processes, we acknowledge that distinguishing between episodic and declarative memory based solely on this approach is not straightforward. We will revise the Supplementary Materials to explicitly discuss these limitations and clarify that our findings do not isolate specific cognitive processes but rather suggest general associations with memory-related networks.

Discussion, Page 17-18 Lines 323-332

“To explore functional relevance, we employed an open-ended cognitive state decoding approach using meta-analytic data (NeuroSynth: Yarkoni et al. (2011)). Although this method usefully generates hypotheses about potential cognitive processes, particularly in the absence of a pre- and post-sleep memory task, it is inherently indirect. Many cognitive terms showed significant associations (16 of 50), such as “episodic memory,” “declarative memory,” and “working memory.” We focused on episodic/declarative memory given the known link with hippocampal reactivation (Diekelmann & Born, 2010; Staresina et al., 2015; Staresina et al., 2023). Nonetheless, these inferences regarding memory reactivation should be interpreted cautiously without direct behavioral measures. Future research incorporating explicit tasks before and after sleep would more rigorously validate these potential functional claims.”

(3) The coupling strength is somehow inconsistent with prior results (Hahn et al., 2020, eLife, Helfrich et al., 2018, Neuron). Specifically, Helfrich et al. showed that among young adults, the spindle is coupled to the peak of the SO. Here, the authors reported that the spindles were coupled to down-to-up transitions of SO and before the SO peak. It is possible that participants' age may influence the coupling (see Helfrich et al., 2018). Please discuss the findings in the context of previous research on SO-spindle coupling.

We appreciate your concern regarding the temporal characteristics of SO-spindle coupling. We acknowledge that the SO-spindle coupling phase results in our study are not identical to those reported by Hahn et al. (2020); Helfrich et al. (2018). However, these differences may arise due to slight variations in event detection parameters, which can influence the precise phase estimation of coupling. Notably, Hahn et al. (2020) also reported slight discrepancies in their group-level coupling phase results, highlighting that methodological differences can contribute to variability across studies. Furthermore, our findings are consistent with those of Schreiner et al. (2021), further supporting the robustness of our observations.

That said, we acknowledge that our original description of SO-spindle coupling as occurring at the “transition from the lower state to the upper state” was not entirely precise. The $-\pi/2$ phase represents the true transition point, while our observed coupling phase is actually closer to the SO peak rather than strictly at the transition. We will revise this statement in the manuscript to ensure clarity and accuracy in describing the coupling phase.

Discussion, Page 16 Lines 283-291

“Our data provide insights into the neurobiological underpinnings of these sleep rhythms. SOs, originating mainly in neocortical areas such as the mPFC, alternate between DOWN- and UP-states. The thalamus generates sleep spindles, which in turn couple with SOs. Our finding that spindle peaks consistently occurred slightly before the UP-state peak of SOs (in 83 out of 107 participants), concurs with prior studies, including Schreiner et al. (2021). Yet it differs from some results suggesting spindles might peak right at the SO UP-state (Hahn et al., 2020; Helfrich et al., 2018). Such discrepancies could arise from differences in detection algorithms, participant age (Helfrich et al., 2018), or subtle variations in cortical-thalamic timing. Nonetheless, these results underscore the importance of coordinated SO-spindle interplay in supporting sleep-dependent processes.”

(4) The discussion is rather superficial with only two pages, without delving into many important arguments regarding the possible functional significance of these results. For example, the author wrote, "This internal processing contrasts with the brain patterns associated with external tasks, such as working memory." Without any references to working memory, and without delineating why WM is considered as an external task even working memory operations can be internal. Similarly, for the interesting results on SO and reduced DMN activity, the authors wrote "The DMN is typically active during wakeful rest and is associated with self-referential processes like mind-wandering, daydreaming, and task representation (Yeshurun, Nguyen, & Hasson, 2021). Its reduced activity during SOs may signal a shift towards endogenous processes such as memory consolidation." This argument is flawed. DMN is active during self-referential processing and mind-wandering, i.e., when the brain shifts from external stimuli processing to internal mental processing. During sleep, endogenous memory reactivation and consolidation are also part of the internal mental processing given the lack of external environmental stimulation. So why during SO or during memory consolidation, the DMN activity would be reduced? Were there differences in DMN activity between SO and SO-spindle coupling events?

We appreciate your concerns regarding the brevity of the discussion and the need for clearer theoretical arguments. We will expand this section to provide more in-depth interpretations of our findings in the context of prior literature. Regarding working memory (WM), we acknowledge that our phrasing was ambiguous. We will modify this statement in the Discussion section.

For the SO-related reduction in DMN activity, we recognize the need for a more precise explanation. This reduced DMN activity may reflect large-scale neural inhibition characteristic of the SO trough. The DMN is typically active during internally oriented cognition (e.g., self-referential processing or mind-wandering) and is suppressed during external stimuli processing (Yeshurun, Nguyen, & Hasson, 2021). It is unlikely, however, that this suppression of DMN during SO events is related to a shift from internal cognition to external responses given it is during deep sleep time. Instead, it could be driven by the inherent rhythmic pattern of SOs, which makes it difficult to separate UP- from DOWN-states (the two temporal regressors were highly correlated, and similar brain activation during SOs events was obtained if modelled at the SO peak instead). Since the amplitude at the SO trough is consistently larger than that at the SO peak, the neural activation we detected may primarily capture the large-scale inhibition from DOWN-state.

To address your final question, we have conducted the additional post hoc comparison of DMN activity between isolated SOs and SO-spindle coupling events. Our results indicate that

DMN activation during SOs was significantly lower than during SO-spindle coupling ($t(106) = -4.17, p < 1e-4$). This suggests that SO-spindle coupling may involve distinct neural dynamics that partially re-engage DMN-related processes, possibly reflecting memory-related reactivation. We appreciate your constructive feedback and will integrate these expanded analyses and discussions into our revised manuscript.

Results, Page 11 Lines 199-208

"Spindles were correlated with positive activation in the thalamus (ROI analysis, $t(106) = 15.39, p < 1e-4$), the anterior cingulate cortex (ACC), and the putamen, alongside deactivation in the DMN (Fig. 3c). Notably, SO-spindle coupling was linked to significant activation in both the thalamus (ROI analysis, $t(106) = 3.38, p = 0.0005$) and the hippocampus (ROI analysis, $t(106) = 2.50, p = 0.0070$, Fig. 3d). However, no decrease in DMN activity was found during SO-spindle coupling, and DMN activity during SO was significantly lower than during coupling (ROI analysis, $t(106) = -4.17, p < 1e-4$). For more detailed activation patterns, see Table S5-S7.

We also varied the threshold used to detect SO events to assess its effect on hippocampal activation during SO-spindle coupling and observed that hippocampal activation remained significant when the percentile thresholds for SO detection ranged between 71% and 80% (see Fig. S6)."

Discussion, Page 17-18 Lines 308-332

"An intriguing aspect of our findings is the reduced DMN activity during SOs when modeled at the SO trough (DOWN-state). This reduced DMN activity may reflect large-scale neural inhibition characteristic of the SO trough. The DMN is typically active during internally oriented cognition (e.g., self-referential processing or mind-wandering) and is suppressed during external stimuli processing (Yeshurun, Nguyen, & Hasson, 2021). It is unlikely, however, that this suppression of DMN during SO events is related to a shift from internal cognition to external responses given it is during deep sleep time. Instead, it could be driven by the inherent rhythmic pattern of SOs, which makes it difficult to separate UP- from DOWN-states (the two temporal regressors were highly correlated, and similar brain activation during SOs events was obtained if modelled at the SO peak instead, Fig. S5). Since the amplitude at the SO trough is consistently larger than that at the SO peak, the neural activation we detected may primarily capture the large-scale inhibition from DOWN-state. Interestingly, no such DMN reduction was found during SO-spindle coupling, implying that coupling may involve distinct neural dynamics that partially re-engage DMN-related processes, possibly reflecting memory-related reactivation. Future research using high-temporal-resolution techniques like iEEG could clarify these possibilities.

To explore functional relevance, we employed an open-ended cognitive state decoding approach using meta-analytic data (NeuroSynth: Yarkoni et al. (2011)). Although this method usefully generates hypotheses about potential cognitive processes, particularly in the absence of a pre- and post-sleep memory task, it is inherently indirect. Many cognitive terms showed significant associations (16 of 50), such as "episodic memory," "declarative memory," and "working memory." We focused on episodic/declarative memory given the known link with hippocampal reactivation (Diekelmann & Born, 2010; Staresina et al., 2015; Staresina et al., 2023). Nonetheless, these inferences regarding memory reactivation should be interpreted cautiously without direct behavioral measures. Future research incorporating explicit tasks before and after sleep would more rigorously validate these potential functional claims."

Recommendations for the authors:

Reviewing Editor Comment:

The reviewers think that you are working on a relevant and important topic. They are praising the large sample size used in the study. The reviewers are not all in line regarding the overall significance of the findings, but they all agree the paper would strongly benefit from some extra work, as all reviewers raise various critical points that need serious consideration.

We appreciate your recognition of the relevance and importance of our study, as well as your acknowledgment of the large sample size as a strength of our work. We understand that there are differing perspectives regarding the overall significance of our findings, and we value the constructive critiques provided. We are committed to addressing the key concerns raised by all reviewers, including refining our analyses, clarifying our interpretations, and incorporating additional discussions to strengthen the manuscript. Below, we address your specific recommendations and provide responses to each point you raised to ensure our methods and results are as transparent and comprehensible as possible. We believe that these revisions will significantly enhance the rigor and impact of our study, and we sincerely appreciate your thoughtful feedback in helping us improve our work.

Reviewer #1 (Recommendations for the authors):

(1) The phrase "overnight sleep" suggests an entire night, while these were rather "nocturnal naps". Please rephrase.

Response: Thank you for pointing this out. We have revised the phrasing in our manuscript to "nocturnal naps" instead of "overnight sleep" to more accurately reflect the duration of the sleep recordings.

(2) Sleep staging results (macroscopic sleep architecture) should be provided in more detail (at least min and % of the different sleep stages, sleep onset latency, total sleep duration, total recording duration), at least mean/SD/range.

Thank you for this suggestion. We will provide comprehensive tables in the supplementary materials, contains descriptive information about sleep-related characteristics. This information will help provide a clearer overview of the macroscopic sleep architecture in our dataset.

Reviewer #2 (Recommendations for the authors):

In order to allow for a better estimation of the reliability of the detected sleep events, please:

(1) Provide densities and absolute numbers of all detected SOs and spindles (N1, NREM, and REM sleep).

Thank you for pointing this out. We will provide comprehensive tables in the supplementary materials, contains detailed information about sleep waves at each sleep stage for all 107 subjects (Table S2-S4), listing for each subject: 1) Different sleep stage duration; 2) Number of detected SOs; 3) Number of detected spindles; 4) Number of detected SO-spindle coupling events; 5) Density of detected SOs; 6) Density of detected spindles; 7) Density of detected SO-spindle coupling events.

Supplementary Materials, Page 43-54, Table S2-S4

(2) Show ERPs for all detected SOs and spindles (per sleep stage).

Thank you for the suggestion. We will provide ERPs for all detected SOs and spindles, separated by sleep stage (N1, N2&N3, and REM) in supplementary Fig. S2-S4. These ERP waveforms will help illustrate the characteristic temporal profiles of SOs and spindles across different sleep stages.

Methods, Page 25, Line 525-532

"Event-related potentials (ERP) analysis. After completing the detection of each sleep rhythm event, we performed ERP analyses for SOs, spindles, and coupling events in different sleep stages. Specifically, for SO events, we took the trough of the DOWN-state of each SO as the zero-time point, then extracted data in a [-2 s to 2 s] window from the broadband (0.1–30 Hz) EEG and used [-2 s to -0.5 s] for baseline correction; the results were then averaged across 107 subjects (see Fig. S2a). For spindle events, we used the peak of each spindle as the zero-time point and applied the same data extraction window and baseline correction before averaging across 107 subjects (see Fig. S2b). Finally, for SO-spindle coupling events, we followed the same procedure used for SO events (see Fig. 2a, Figs. S3–S4)."

(3) Provide detailed info concerning sleep characteristics (time spent in each sleep stage etc.).

Thank you for this suggestion. Same as the response above, we will provide comprehensive tables in the supplementary materials, contains descriptive information about sleep-related characteristics.

Supplementary Materials, Page 42, Table S1 (same as above)

(4) What would happen if more stringent parameters were used for event detection? Would the authors still observe a significant number of SO spindles during N1 and REM? Would this affect the fMRI-related results?

Thank you for this suggestion. Our methods for detecting SOs, spindles, and their couplings were originally developed for N2 and N3 sleep data, based on the specific characteristics of these stages. These methods are widely recognized in sleep research (Hahn et al., 2020; Helfrich et al., 2019; Helfrich et al., 2018; Ngo, Fell, & Staresina, 2020; Schreiner et al., 2022; Schreiner et al., 2021; Staresina et al., 2015; Staresina et al., 2023). However, because this percentile-based detection approach will inherently identify a certain number of events if applied to other stages (e.g., N1 and REM), the nature of these events in those stages remains unclear compared to N2/N3. We nevertheless identified and reported the detailed descriptive statistics of these sleep rhythms in all sleep stages, under the same operational definitions, both for completeness and as a sanity check. Within the same subject, there should be more SOs, spindles, and their couplings in N2/N3 than in N1 or REM (see also Figure S2-S4, Table S1-S4).

Furthermore, in order to explore the impact of this on our fMRI results, we conducted an additional sensitivity analysis by applying different detection parameters for SOs. Specifically, we adjusted amplitude percentile thresholds for SO detection (the parameter that has the greatest impact on the results). We used the hippocampal activation value during N2&N3 stage SO-spindle coupling as an anchor value and found that when the parameters gradually became stricter, the results were similar to or even better than the current results. However, when we continued to increase the threshold, the results began to gradually decrease until the threshold was increased to 80%, and the results were no longer significant. This indicates that our results are robust within a specific range of parameters, but as the threshold increases, the number of trials decreases, ultimately weakening the statistical power of the fMRI analysis.

Thank you again for your suggestions on sleep rhythm event detection. We will add the results in Supplementary and revise our manuscript accordingly.

Results, Page 11, Line 199-208

“Spindles were correlated with positive activation in the thalamus (ROI analysis, $t(106) = 15.39$, $p < 1e-4$), the anterior cingulate cortex (ACC), and the putamen, alongside deactivation in the DMN (Fig. 3c). Notably, SO-spindle coupling was linked to significant activation in both the thalamus (ROI analysis, $t(106) = 3.38$, $p = 0.0005$) and the hippocampus (ROI analysis, $t(106) = 2.50$, $p = 0.0070$, Fig. 3d). However, no decrease in DMN activity was found during SO-spindle coupling, and DMN activity during SO was significantly lower than during coupling (ROI analysis, $t(106) = -4.17$, $p < 1e-4$). For more detailed activation patterns, see Table S5-S7. We also varied the threshold used to detect SO events to assess its effect on hippocampal activation during SO-spindle coupling and observed that hippocampal activation remained significant when the percentile thresholds for SO detection ranged between 71% and 80% (see Fig. S6).”

Finally, we sincerely thank all again for your thoughtful and constructive feedback. Your insights have been invaluable in refining our analyses, strengthening our interpretations, and improving the clarity and rigor of our manuscript. We appreciate the time and effort you

have dedicated to reviewing our work, and we are grateful for the opportunity to enhance our study based on your recommendations.

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